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Design mechanisms and biomedical applications of peptide-metal chelates in antimicrobial therapy, tumor theranostics, and integrated diagnosis-treatment systems ☆

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Outline

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Highlights

- Summarises core design principles and mechanisms of peptide-metal chelates.
- Elaborates on the “Trojan horse” strategy against antimicrobial resistance.
- Highlights the biomedical applications of peptide-metal chelates
- Proposes AI-assisted design and multimodal integration solutions.

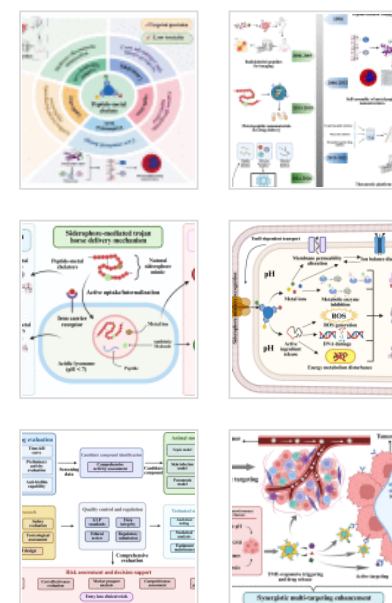
Abstract

The growing global antimicrobial resistance crisis and limitations of conventional cancer therapies call for innovative biomedical strategies. Peptide-metal chelates are promising multifunctional biomaterials that leverage peptide-metal ion synergy to overcome traditional therapeutic bottlenecks, with core value in building integrated platforms for programmable targeted delivery, spatiotemporal smart responsiveness and intrinsic theranostic synergy. Research in this field has evolved from basic molecular discovery to systematic rational design and now clinical smart applications. This review presents a novel research and development roadmap for peptide-metal chelates, elaborating their design principles, structure-function mechanisms and latest biomedical advances. It highlights their unique “Trojan horse” strategy for antibacterial resistance and precise tumor targeting *via* the EPR effect and tumor microenvironmental triggers such as pH and enzymes. It details their applications in intelligent drug delivery, high-efficacy antimicrobial therapy, precision anticancer treatment, and theranostic platforms integrating imaging and therapy. Addressing gaps in existing fragmented summaries, including the lack of systematic design-synthesis-application integration and insufficient basic-clinical translation analysis, the review also notes unresolved challenges in long-term *in vivo* safety, bioavailability optimization and GMP-compliant large-scale production. Finally, it prospects core directions like AI-assisted rational molecular design, advanced multi-stimuli responsive materials and multimodal theranostic integration, which are expected to accelerate the clinical translation of peptide-metal chelates and offer innovative solutions for drug-resistant infections and refractory cancers.

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Keywords

Peptide-metal chelates; Antimicrobial therapy; Tumor theranostics; Drug delivery system; Integrated diagnosis and therapy; Clinical translation

Abbreviations

AFM, atomic force microscopy; AI, artificial intelligence; ALP, alkaline phosphatase; ALT, alanine aminotransferase; AST, aspartate aminotransferase; ATP, adenosine triphosphate; ATR, active transport and retention; AUC, area under the curve; BBB, blood-brain barrier; BCMA, B-cell maturation antigen; BSA, bovine serum albumin; CAFs, cancer-associated fibroblasts; CD, circular dichroism; CD13, cluster of differentiation 13; CD38, cluster of differentiation 38; CD44, cluster of differentiation 44; CDT, chemodynamic therapy; c-Met, hepatocyte growth factor receptor; COSY, correlation spectroscopy; CPP, cell-penetrating peptide; CT, computed tomography; CYP, cytochrome P450; DAMP, damage-associated molecular pattern; DMF, dimethylformamide; DNA, deoxyribonucleic acid; DOTA, 1,4,7,10-tetraazacyclododecane-1,4,7,10-tetraacetic acid; d-d transition, d-d electron transition; EC₅₀, half maximal effective concentration; EDB-FN, extracellular matrix B fibronectin; EPR, enhanced permeability and retention; ESI-MS, electrospray ionization

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mass spectrometry; EUCAST, European Committee on Antimicrobial Susceptibility Testing; FAP, fibroblast activation protein; Fmoc, fluorenylmethyloxycarbonyl; FRET, fluorescence resonance energy transfer; GANs, generative adversarial networks; GAPDH, glyceraldehyde-3-phosphate dehydrogenase; GBCA, gadolinium-based contrast agent; GGT, gamma-glutamyl transferase; GLP, good laboratory practice; GSH, glutathione; HA, hyaluronic acid; HAase, hyaluronidase; HER2, human epidermal growth factor receptor 2; His, histidine; HPLC, high-performance liquid chromatography; HSP70, heat shock protein 70; ICD, immunogenic cell death; IC₅₀, half maximal inhibitory concentration; ID, injected dose; IL, interleukin; iRGD, internalizing arginine-glycine-aspartic acid peptide; LA-ICP-MS, laser ablation-inductively coupled plasma mass spectrometry; LC-MS, liquid chromatography-mass spectrometry; LD₅₀, median lethal dose; LDH, lactate dehydrogenase; LHRH, luteinizing hormone-releasing hormone; LN, lipid nanoparticles; LPS, lipopolysaccharide; MALDI-TOF MS, matrix-assisted laser desorption/ionization time-of-flight mass spectrometry; MBC, minimum bactericidal concentration; MBEC, minimum biofilm eradication concentration; MCP-1, monocyte chemoattractant protein-1; MECAM, microbial extracellular matrix adhesion molecule; MIC, minimum inhibitory concentration; MLCT, metal-to-ligand charge transfer; MMP, matrix metalloproteinase; MOF, metal-organic framework; MRI, magnetic resonance imaging; mRNA, messenger ribonucleic acid; MS, mass spectrometry; NF- κ B, nuclear factor kappa-light-chain-enhancer of activated B cells; NIR, near-infrared; NIRF, near-infrared fluorescence; NMR, nuclear magnetic resonance; NOAEL, no-observed-adverse-effect level; NOESY, nuclear Overhauser effect spectroscopy; NOTA, 1,4,7-triazacyclononane-1,4,7-triacetic acid; NSF, nephrogenic systemic fibrosis; PBS, phosphate-buffered saline; PCY1, prolyl cyclase 1; PD-1, programmed cell death protein 1; PDT, photodynamic therapy; PEG, polyethylene glycol; PET, positron emission tomography; pK_a, acid dissociation constant; PMO, phosphorodiamidate morpholino oligomer; PROTAC, proteolysis targeting chimera; PSMA, prostate-specific membrane antigen; RES, reticuloendothelial system; RiPPs, ribosomally synthesized and post-translationally modified peptides; RNA, ribonucleic acid; ROS, reactive oxygen species; SBR, signal-to-background ratio; SDT, sonodynamic therapy; SEM, scanning electron microscopy; siRNA, small interfering ribonucleic acid;

SPECT, single-photon emission computed tomography; SPPS, solid-phase peptide synthesis; SUVmax, maximum standardized uptake value; T_1 , longitudinal relaxation time; tBu, tert-butyl; TEM, transmission electron microscopy; TLR, toll-like receptor; TME, tumor microenvironment; TNF- α , tumor necrosis factor-alpha; UV-Vis, ultraviolet-visible; WHO, World Health Organization; XANES, X-ray absorption near-edge structure spectroscopy; ^{64}Cu , copper-64; ^{68}Ga , gallium-68; $^{99\text{m}}\text{Tc}$, technetium-99m; ^{177}Lu , lutetium-177.

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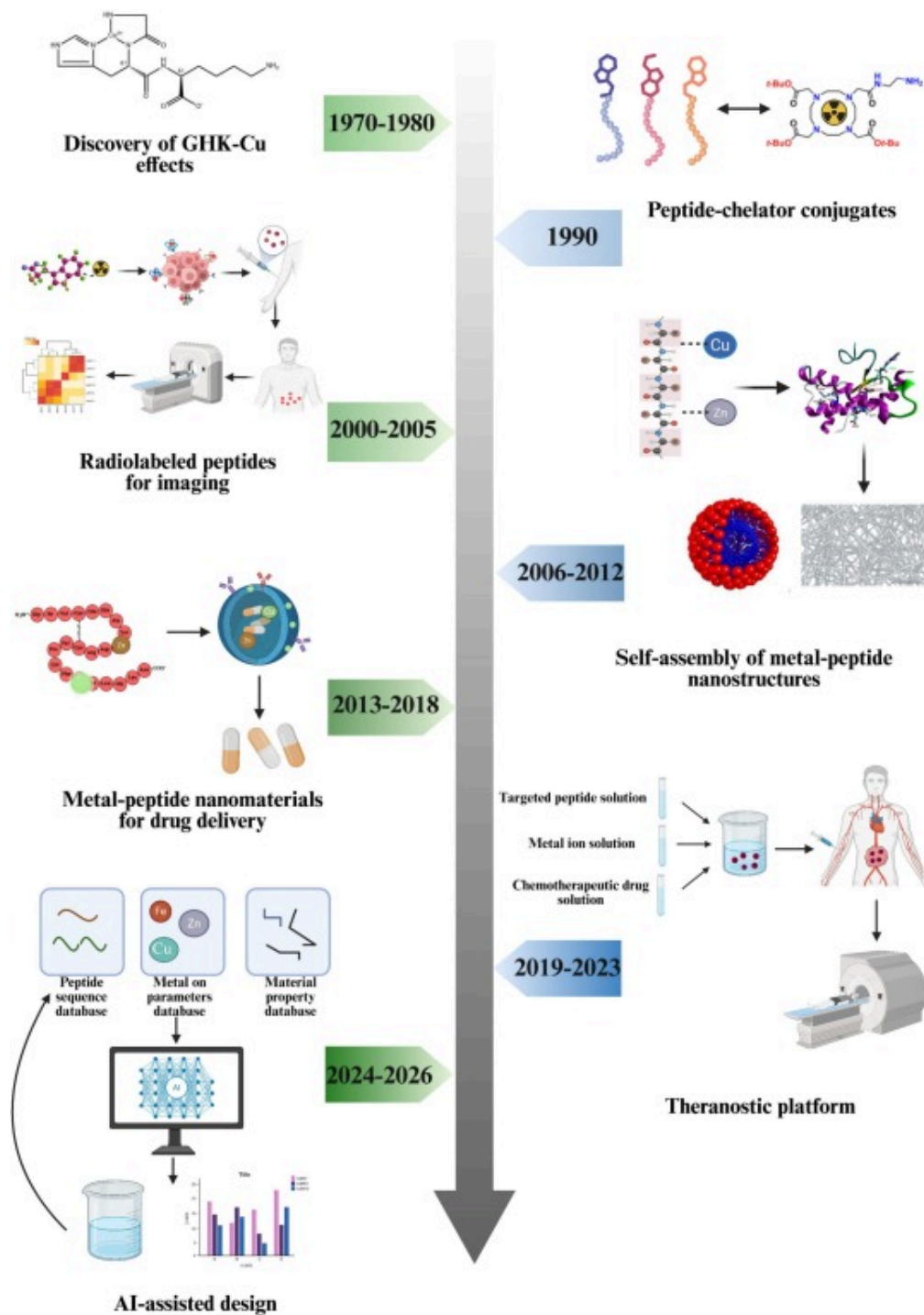
1. Introduction

Antimicrobial resistance, particularly infections caused by multidrug-resistant bacteria, has been widely recognized as a major crisis in global public health. The field of cancer treatment also faces significant challenges, as traditional methods such as surgery, radiation therapy, and chemotherapy have clear limitations. These limitations are manifested respectively in the difficulty of surgery to eliminate metastatic cancer cells, the non-selective attack of chemotherapy on healthy cells causing side effects like weakened immunity, anemia, and hair loss, and the potential of radiation therapy to damage surrounding healthy tissues. [1]. Against this backdrop, peptide-metal chelates emerge as novel biomedical materials whose core advantage lies in their ability to construct multifunctional integrated platforms. On the one hand, by mimicking natural pathways such as bacterial iron carriers, they achieve “Trojan horse”-style targeted delivery (where nanoparticles are internalized by cells to deliver therapeutics from within), overcoming biological barriers [2]. On the other hand, leveraging the coordination interactions between metal ions and peptides not only enhances the stability and activity of existing drugs but may also endow them with novel therapeutic mechanisms, thereby achieving synergistic enhancement and reduced toxicity [3].

The design and functional realization of peptide-metal chelates are based on theoretical foundations such as coordination chemistry and molecular self-assembly [4]. Amino acid residues such as histidine, cysteine, and aspartic acid in polypeptide chains can act as potent ligands, forming stable coordination bonds with metal ions, iron (Fe), copper (Cu), zinc (Zn), gold (Au), and ruthenium (Ru) [5]. This interaction not only underpins structural stability but also drives controlled self-assembly of molecules, forming diverse ordered higher-order structures such as nanofibers and nanoparticles. These structures provide ideal carriers for efficient drug loading, protection, and controlled release. For instance, amphiphilic peptides can self-assemble into nanofibers with hydrophobic cores, effectively encapsulating hydrophobic metal drugs [6].

In recent years, peptide-metal chelates have achieved remarkable progress in specific applications across multiple biomedical fields. As shown in Fig. 1, research on peptide-metal chelates has evolved from fundamental discoveries to systematic design, ultimately advancing toward clinical smart applications. Early research centered on natural peptide-metal complexes. Subsequently, radiolabeling techniques propelled their use in disease imaging and targeted radiotherapy. In recent years, integration with nanotechnology has enabled the controlled assembly of self-assembling nanostructures and the delivery of intelligent drugs. Currently, the field is moving toward deep integration into integrated diagnosis-treatment platforms, incorporating AI-assisted design, demonstrating a clear trend from single-function to systematic, precision-targeted therapies. For instance, in combating drug-resistant bacterial infections, researchers are developing novel “dual-targeting” metal-peptide antibiotics by conjugating cyclic peptide antibiotics with metal complexes to achieve novel mechanisms of action and enhanced antimicrobial activity. In tumor-targeted therapy, peptide-metal chelates offer diverse, highly promising application strategies. Studies indicate that binding host defense peptides Pexidin 1 and 3 to copper ions can increase their cytotoxicity against cancer cells by up to 2-fold and 7-fold, respectively [7]. Beyond directly enhancing peptide activity, peptide-metal assemblies have also been engineered as smart drug delivery systems. For instance, amphiphilic peptides responsive to matrix metalloproteinase-9 (MMP-9), which is

overexpressed in the tumor microenvironment, successfully encapsulated hydrophobic gold-based anticancer drugs to form nanofibers. This system enables specific enzymatic cleavage and drug release at tumor sites, demonstrating greater cytotoxicity than free drugs in renal and breast cancer cell lines while remaining harmless to normal lung fibroblasts, thus exhibiting excellent selectivity [6]. Moreover, in the field of integrated diagnosis and therapy, peptide-metal chelates have emerged as ideal materials for constructing integrated platforms that combine precise diagnostic capabilities with efficient therapeutic functions, owing to their unique chemical designability and biocompatibility. The core advantage of such systems is the true multifunctional integration within a single entity. This integration combines the peptide's inherent targeting and environmental responsiveness with the metal ion's imaging capabilities, including radiographic, fluorescent, photoacoustic, and magnetic resonance signals, as well as therapeutic activity. For example, linking a peptide targeting CD38 on multiple myeloma cell surfaces to the radionuclide copper-64 *via* a chelating agent directly constructs a targeted probe for specific cancer diagnostic imaging, providing critical information for subsequent therapeutic interventions [8].



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Fig. 1. The evolution and key milestones in peptide-metal chelate research and development.

This paper provides a systematic review of the design principles, functional mechanisms, and application progress of peptide-metal chelates in the biomedical field. It focuses on analyzing their unique advantages in combating drug-resistant bacterial infections and in targeted tumor therapy, while exploring strategies to construct them as smart drug delivery systems and integrated diagnosis-treatment platforms. Despite significant application potential, this field still faces critical challenges, including evolving drug resistance, bioavailability limitations, and long-term safety concerns. Looking ahead, the integration of cutting-edge technologies, such as AI-assisted molecular design, smart responsive materials, and multimodal diagnostic-therapeutic systems, promises to advance peptide-metal chelates toward greater precision and intelligence. This evolution is expected to accelerate their clinical translation, offering innovative solutions to major therapeutic challenges in disease management.

2. Structural design and synthesis methods of peptide-metal chelating complexes

2.1. Peptide sequence design strategies

Rational design of peptide sequences is central to realizing the functionality of peptide-metal chelates. Design strategies must comprehensively consider the selection of functional amino acids, the optimization of metal-binding sites, the introduction of targeted ligands, and the balance between stability and biocompatibility. This design process relies on the precise utilization of various functional amino acids. As shown in [Table 1](#), metal-binding amino acids coordinate with specific metal ions through their side-chain groups to form active sites or stabilize structures; targeted recognition amino acids

mediate cell penetration, molecular recognition, and specific binding; while structure-stabilizing amino acids optimize the peptide backbone through conformational restriction or flexibility provision. Selection criteria for functional peptides primarily consider factors such as affinity for specific biological targets, cell-penetrating capability, and *in vivo* stability [9]. Research indicates that histidine-containing peptide sequences serve as key sites for chemically selective modification. Their imidazole side chains can undergo specific interactions with heterophase copper catalysts, providing a basis for activating and achieving site-selective functionalization of peptides [10].

Table 1. Common functional amino acids and their uses in peptide design.

Amino acid type	Representative amino acid	Functional characteristics	Application field	Ref.
Metal-binding amino acids	Histidine (His)	The imidazole group provides coordination sites, forming stable complexes with Fe, Cu, Zn, <i>etc.</i>	Metal enzyme mimicry, catalytic center construction, and metal drug targeting.	[11]
	Cysteine (Cys)	The thiol group has a high affinity for soft acid metals (<i>e.g.</i> , Hg ²⁺ , Cd ²⁺) and can form disulfide bonds.	Heavy metal chelation, structural stabilization, and covalent inhibitor design.	[12]
	Aspartic acid (Asp) / Glutamic acid (Glu)	Side-chain carboxyl groups provide oxygen atom coordination sites, preferentially binding hard acid metals like Ca ²⁺ , Mg ²⁺ .	Calcium signaling, bone-targeting materials, mineral-binding peptides.	[13]

Amino acid type	Representative amino acid	Functional characteristics	Application field	Ref.
Targeting/Recognition amino acids	Methionine (Met)	The sulfur atom of the thioether group can serve as a coordination site for soft metals (<i>e.g.</i> , Pt ²⁺ , Ru ²⁺).	Metal antitumor drug design, metalloprotein engineering.	[14]
	Arginine (Arg)	Positively charged guanidinium group interacts with negatively charged cell membrane surfaces, enhancing cell penetration.	Cell-penetrating peptides (CPP), nuclear localization signals (NLS).	[15]
	Asparagine (Asn)	The amide group is a key site for N-linked glycosylation and is involved in specific sugar recognition.	Glycosylation recognition, bacterial targeting, protein folding & quality control.	[16]
	Tyrosine (Tyr)	The phenolic hydroxyl group can be phosphorylated and is involved in signal transduction.	Receptor binding, phosphorylation signal regulation, and molecular recognition probes.	[17]
	Phenylalanine (Phe) / Tryptophan (Trp)	Bulky, hydrophobic aromatic side chains participate in hydrophobic interactions, π - π stacking, and cation- π interactions.	Protein-protein interaction interfaces, targeting hydrophobic pockets, and	[18], [19]

Amino acid type	Representative amino acid	Functional characteristics	Application field	Ref.
Structure-stabilizing amino acids			membrane protein anchoring.	
	Lysine (Lys)	Positively charged ϵ -amino group can bind to a negatively charged. Groups, and is also a site for various post-translational modifications.	Enzyme substrate recognition, protein interaction modulation, and peptide conjugation site.	[20]
	Proline (Pro)	Rigid cyclic structure, which induces β -turns or terminates α -helices, restricts conformational flexibility.	Conformational constraint, stability enhancement, turn construction.	[21]
	Glycine (Gly)	The side chain is only a hydrogen atom, provides maximal backbone conformational flexibility, and promotes turn formation.	Chain flexibility regulation, turn construction, linker design.	[22]
	Alanine (Ala)	Simple methyl hydrophobic side chain, often used in mutation studies to replace other amino acids to probe side-	Hydrophobic core packing, helix formation promotion, and mutation studies.	[23]

Amino acid type	Representative amino acid	Functional characteristics	Application field	Ref.
		chain function, also has a mild hydrophobic effect.		
	Valine (Val) / Isoleucine (Ile) / Leucine (Leu)	Hydrophobic side chains with β -branching favour β -sheet formation strongly and contribute to hydrophobic core packing.	β -sheet structure formation, hydrophobic core stabilization, protein aggregation inhibition.	[24]

The design of metal-binding sites requires precise control over coordination geometry and binding strength. Cysteine-rich peptides such as metallothionein-3 can effectively chelate metal ions. For instance, this protein coordinates with Bi^{3+} *via* the thiol groups of its cysteine residues, synergistically forming a unique Bi_2S_5 cluster structure characterized by shared thiolate ligands [25]. Research has also revealed that both CX3C and CX2C, two cysteine-containing curled-helix peptides, exhibit strong binding affinity for heavy metal ions such as $\text{Cd}(\text{II})$, $\text{Pb}(\text{II})$, and $\text{Hg}(\text{II})$. However, significant differences exist between them in terms of binding affinity and coordination geometry [26].

Strategies for introducing targeted ligands include integrating tumor-specific recognition sequences, bacterial iron transporter mimic sequences, or cell-penetrating peptides into the peptide backbone. The rational arrangement of these functional modules enables precise drug delivery and cellular internalization. Stability optimization enhances protease resistance and *in vivo* half-life through amino acid substitutions, N-terminal acetylation, C-terminal amidation, and the incorporation of non-natural amino acids [27].

Optimizing biocompatibility involves reducing immunogenicity, lowering toxicity, and improving pharmacokinetic properties. By introducing D-amino acids, polyethylene glycol modifications, or lipidation, the *in vivo* stability and bioavailability of peptides can be

significantly enhanced [28], [29]. Research indicates that rational peptide design enables metal-induced folding to overcome the natural unfolding tendency of polar cysteine residues, thereby achieving a balance between functionality and stability [26].

2.2. Selection and coordination properties of metal ions

Based on an in-depth understanding of the characteristics of different metal ions, the design of peptide-metal chelates has been highly rationalized and functionally oriented. As shown in Table 2, the design strategies for different metal ions exhibit distinct functional differentiation. Regarding oxidative stress-mediated therapy, redox-active metal ions have garnered widespread attention. Copper ions ($\text{Cu}^+/\text{Cu}^{2+}$), which can catalyze Fenton-like reactions to generate reactive oxygen species, have been widely used in chemodynamic therapy and antibacterial research. Meanwhile, cobalt ($\text{Co}^{2+}/\text{Co}^{3+}$) and ruthenium ($\text{Ru}^{2+}/\text{Ru}^{3+}$), leveraging their rich photochemical properties, demonstrate unique advantages in photodynamic/photothermal therapy. Turning to imaging diagnosis and theranostics, lanthanides and radioactive metals exhibit particularly pronounced advantages. Gadolinium ions (Gd^{3+}), by virtue of their strong paramagnetism that significantly shortens the T_1 relaxation time of water protons, have become the gold standard for MRI T_1 -weighted contrast agents. Notably, copper-64 (^{64}Cu), which combines the dual functions of emitting β^+ particles (for PET imaging) and β^- particles (for therapy), along with its favorable half-life of 12.7h, positions it as an ideal candidate for the development of theranostic probes. From a biomimetic perspective, certain metals embody unique bio-inspired strategies. For instance, gallium ions (Ga^{3+}) exhibit coordination modes similar to those of iron ions but lack redox activity, enabling them to exert antibacterial effects by interfering with iron-dependent metabolism. This makes them important carriers for the “Trojan horse” antibacterial strategy.

Table 2. Comparison of characteristics, functions, and application fields of different metal ions.

Metal ion	Coordination geometry	Preferred coordinating atoms	Main biological function/characteristic	Main application field
Copper (Cu ⁺ /Cu ²⁺)	Tetrahedral (Cu ⁺)/ Square pyramidal or Octahedral (Cu ²⁺).	N (His)、S (Met, Cys)、 O (Asp)	Redox catalysis, reactive oxygen species (ROS) generation.	Chemodynamic therap (CDT), oxidative stress- mediated antibacterial imaging probe (⁶⁴ Cu).
Zinc (Zn ²⁺)	Tetrahedral/Octahedral.	S (Cys)、N (His)、O (Asp, Glu)	Structural stabilization, Lewis acid catalysis.	Artificial metalloenzym mimicry, gene regulati tools, and antibacterial
Gallium (Ga ³⁺)	Octahedral.	O (carboxylate, N phenolate), N (imidazole)	Iron mimetic, lacks redox activity, interferes with iron- dependent metabolism.	“Trojan horse”-style antibacterial therapy, antitumor.
Gadolinium (Gd ³⁺)	Octahedral/Nine- coordinate.	O (carboxylate, N phosphate), N (amide)	Strong paramagnetism significantly shortens the water proton T1 relaxation time.	MRI T1-weighted contr agent, theranostics.
Technetium (^{99m} Tc)	Square pyramidal/Octahedral.	S (thiol), N (amide), O (carboxylate)	Emits 140keV γ-rays, suitable for imaging.	SPECT/CT imaging.
Lutetium (¹⁷⁷ Lu)	Octahedral.	O (carboxylate, N DOTA), N (amide), S (thiol)	Emits β ⁻ particles and has cell-killing ability.	Peptide receptor radiot therapy, radioimmunotherapy.

Metal ion	Coordination geometry	Preferred coordinating atoms	Main biological function/characteristic	Main application field
Copper-64 (^{64}Cu)	Square pyramidal/Octahedral.	N (His), S (Cys), O (carboxylate, DOTA)	DOTA)Emits β^+ particles (PET) and β^- particles (therapy), half-life 12.7h.	PET imaging, theranost (paired with ^{177}Lu).
Manganese (Mn^{2+})	Octahedral/Square pyramidal.	O (carboxylate, phosphate), N (His)	Paramagnetic species can serve as cofactors for enzymes such as MnSOD.	MRI T1 contrast agent (specific), enzyme-mim antioxidant.
Cobalt ($\text{Co}^{2+}/\text{Co}^{3+}$)	Octahedral/Tetrahedral.	N (His), S (Cys), O (carboxylate)	Redox-active, involved in the vitamin B12 coenzyme.	Anticancer drugs, catal materials.
Ruthenium ($\text{Ru}^{2+}/\text{Ru}^{3+}$)	Octahedral.	N (imidazole, aromatic N of ligands), S (Met)	Photoactivation generates reactive species, and transfer hydrogenation catalysis.	Photodynamic/photoc therapy, anticancer dru and artificial photosyn
Platinum (Pt^{2+})	Square planar.	N (guanine, His), S (Met)	Crosslinks DNA, blocks replication and transcription.	Classical anticancer chemotherapeutic dru (cisplatin, oxaliplatin).
Gold ($\text{Au}^+/\text{Au}^{3+}$)	Linear (Au^+)/ Square planar (Au^{3+}).	S (Cys), P (phosphine ligands), N (ligands)	N (ligands)inhibit thioredoxin reductase, anti-inflammatory.	Anti-rheumatic drugs (auranofin), anticancer research, and catalysis.

Metal ion	Coordination geometry	Preferred coordinating atoms	Main biological function/characteristic	Main application field
Calcium (Ca ²⁺)	Six- to eight-coordinate, irregular.	O (carboxylate-Asp/Glu, carbonyl, phosphate)	Universal secondary messenger, triggers muscle contraction, neurotransmission, <i>etc.</i>	Bone tissue engineering, biomineralization regulation and signaling probes.
Magnesium (Mg ²⁺)	Octahedral.	O (phosphate-ATP, carboxylate, water)	Essential for ATPase catalysis, stabilizes nucleic acid structures, and is the core of chlorophyll.	Enzyme cofactor mimic, nucleic acid stabilization, biomaterials.

The correlation between these coordination characteristics and biological functions suggests that precise modulation of the metal center's coordination environment holds promise for expanding functionality from monotherapy to multimodal theranostics.

Iron ions (Fe²⁺/Fe³⁺) play a central role in bacterial iron uptake systems. Their chelates can effectively mimic the function of natural iron carriers, enabling targeted drug delivery *via* bacterial-specific transport systems [48]. Research indicates that in aqueous solutions, Fe(II) ions can form stable binary and ternary complexes with various amino acids, including histidine, as well as ligands such as pyridinecarboxylic acid [49]. The formation patterns and stability studies of such small-molecule complexes provide crucial insights into the chemical basis of amino acid side-chain chelation of metal ions in peptides.

Copper ions (Cu⁺/Cu²⁺) exhibit potent redox activity and can be used in chemotherapeutic therapy. Cu²⁺ catalyzes the Fenton-like reaction, converting hydrogen peroxide into highly

toxic hydroxyl radicals, thereby inducing tumor cell apoptosis [50]. Copper ions typically exhibit tetrahedral or octahedral coordination geometries, forming stable complexes with ligands such as histidine, cysteine, and asparagine [51]. Zinc ions (Zn^{2+}) play a crucial role in regulating enzyme activity, acting as Lewis acids to catalyze hydrolysis reactions or stabilizing protein structures. Zinc ions prefer tetrahedral coordination environments, forming stable zinc finger structures with cysteine and histidine residues [52].

Gallium ions (Ga^{3+}) can replace iron and interfere with bacterial metabolic processes because of their similar ionic radius and coordination chemistry to Fe^{3+} . While lacking redox activity, their high charge density confers an affinity comparable to that of iron carriers, enabling them to inhibit bacterial growth through a “Trojan horse” strategy [53]. Radioactive metals such as technetium-99m (^{99}Tc), lutetium-177 (^{177}Lu), and copper-64 (^{64}Cu) hold unique value in diagnostic and therapeutic applications. Among these, ^{64}Cu 's half-life and decay characteristics make it an ideal isotope for both PET imaging and therapy [54].

The coordination chemistry characteristics determine the interaction patterns between metal ions and peptides, thereby influencing the stability and functional performance of the chelates. Hard-acid metal ions such as Fe^{3+} and Ga^{3+} tend to bind to hard-base ligands, such as carboxylate and phenolate oxygen atoms. In contrast, soft-acid metal ions such as Cu^+ and Hg^{2+} prefer soft-base ligands, such as thiol groups and imidazole nitrogen [55]. This coordination selectivity provides the theoretical basis for rationally designing high-affinity chelating agents for specific metal ions.

2.3. Synthesis methods and characterization techniques

Various methods exist for synthesizing peptide-metal chelates, among which solid-phase peptide synthesis (SPPS) has emerged as the mainstream approach due to its high efficiency and precision. Its core principle involves covalently anchoring the first amino acid of the target molecule to an insoluble resin carrier using acid-sensitive linkers, followed by repeated cycles of deprotection, coupling, and washing to extend the peptide

chain [56]. The Fmoc/tBu protection strategy is widely adopted due to its mild deprotection conditions (20% piperidine/DMF) and excellent side-chain compatibility, making it particularly suitable for synthesizing peptide chains containing acid-labile amino acids such as tryptophan [57].

Metal-organic frameworks (MOFs) provide a novel synthetic platform for the construction of peptide-metal complexes. MOFs are crystalline porous materials formed by the self-assembly of metal ions or clusters with organic ligands *via* coordinate bonds, exhibiting tunable pore sizes (3–30 Å) and high specific surface areas [58]. Researchers have developed a MOF-catalyzed polypeptide synthesis method that utilizes exposed metal clusters on the surface of MOF nanoparticles as catalytic centers to initiate polymerization. This approach enables the one-step preparation of high-quality polypeptides and MOF-polypeptide composites [59].

The characterization technology system provides comprehensive assurance for quality control and performance evaluation of peptide-metal chelates. UV–Vis spectroscopy provides information on coordination structures and electronic states through characteristic absorption bands, such as the metal-ligand charge-transfer band and the d-d transition band. Circular dichroism (CD) technology analyzes secondary structural changes in proteins and peptides by exploiting the differential absorption of left- and right-handed circularly polarized light by chiral substances [60]. Far-ultraviolet circular dichroism (190–250 nm) reflects the arrangement of peptide bonds, enabling quantitative calculation of the proportions of α -helix, β -sheet, turn, and irregular coil structures [61]. Near-ultraviolet CD (250–320 nm) reveals changes in the microenvironment of aromatic amino acid side chains and disulfide bonds [62].

Nuclear magnetic resonance (NMR) techniques, particularly two-dimensional NMR methods such as COSY and NOESY, provide atomic-level structural information for determining coordination geometry, bond lengths and angles, and molecular conformation [63]. Mass spectrometry (MS) accurately determines molecular weight and elemental composition, with electrospray ionization mass spectrometry (ESI-MS) and

matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI-TOF MS) being commonly used analytical techniques [64]. Transmission electron microscopy (TEM) and scanning electron microscopy (SEM) provide direct visualization of nanoscale morphological features and size distributions. Atomic force microscopy (AFM) also provides surface topography and mechanical property information [65]. The integrated application of these techniques ensures controllability in the composition, structure, purity, and functionality of peptide-metal chelates, laying a solid foundation for subsequent biomedical applications.

2.4. Toxicity mechanisms, monitoring and detection methods, and mitigation strategies for different metal ions

The *in vivo* release and accumulation of metal ions from peptide-metal chelates represent a central safety concern for their clinical translation. Due to differences in electron configuration, redox activity, and metabolic pathways, different metal ions exhibit highly specific toxicity mechanisms, with chelate stability, administration routes, and dosage significantly influencing the resulting toxic effects. This section systematically reviews the toxicity mechanisms of commonly used metal ions, establishes a multi-tiered framework for monitoring and detection, and proposes mitigation strategies encompassing rational design, process regulation, and clinical intervention.

2.4.1. Mechanisms underlying metal ion toxicity

Gadolinium ion (Gd^{3+}) is the most commonly used metal ion for magnetic resonance imaging contrast agents in peptide-metal chelates, and the core of its toxicity lies in accumulative damage *in vivo* [66]. Clinically used linear gadolinium-based contrast agents (GBCAs) are prone to *in vivo* dechelation due to their relatively low coordination stability. The released free Gd^{3+} can circulate *via* the bloodstream and deposit in tissues, including the brain, kidneys, liver, and bones, with the extent of deposition positively correlated with the number of administrations [67]. Brain Gd^{3+} deposition is primarily concentrated in regions such as the dentate nucleus and globus pallidus. Long-term accumulation may

disrupt central nervous system ion homeostasis, thereby potentially affecting neuronal signal transduction [68]. The kidneys serve as the primary metabolic organs for Gd^{3+} clearance. Free Gd^{3+} can accumulate within renal tubular epithelial cells, where it induces oxidative stress and generates excessive reactive oxygen species (ROS), leading to mitochondrial dysfunction in these cells. In severe cases, this can trigger nephrogenic systemic fibrosis (NSF), a condition with significantly elevated risk, particularly in patients with renal insufficiency [69].

Copper ions (Cu^+/Cu^{2+}) serve as central metal ions in chemodynamic therapy (CDT), with their toxicity primarily arising from dysregulated redox activity and perturbation of copper metabolic pathways. Under normal physiological conditions, the uptake, transport, and excretion of copper are tightly regulated by proteins such as ceruloplasmin, ATP7A, and ATP7B. However, excessive free Cu^{2+} released from peptide-metal chelates disrupts this balance. On one hand, Cu^{2+} can non-specifically generate reactive oxygen species (ROS) *via* Fenton or Fenton-like reactions. These ROS not only exert cytotoxic effects on tumor cells but also damage normal cellular components, including DNA, proteins, and lipids, ultimately leading to membrane lipid peroxidation and apoptosis [69]. On the other hand, excessive Cu^{2+} can accumulate in organs such as the liver and spleen, where it interferes with the activity of copper-dependent enzymes, inhibits mitochondrial respiratory chain function, and consequently impairs energy metabolism [70], [71].

Platinum ions (Pt^{2+}) are classic antitumor metal ions, with their toxicity primarily manifested as non-specific DNA damage and cytotoxicity toward normal tissues. Pt^{2+} can form crosslinks with the DNA of healthy cells, thereby blocking DNA replication and transcription. This not only inhibits tumor cell proliferation but also damages rapidly dividing normal cells, including bone marrow hematopoietic cells, gastrointestinal mucosal cells, and germ cells, leading to typical adverse effects such as myelosuppression, nausea and vomiting, alopecia, and reproductive toxicity. Furthermore, platinum ions can accumulate in the kidneys, impairing tubular reabsorption function and inducing

nephrotoxicity. Additionally, they may interfere with the metabolism of auditory nerve cells, resulting in ototoxicity that clinically presents as hearing loss and tinnitus [72].

Gallium ions (Ga^{3+}) exert antimicrobial and antitumor effects by mimicking Fe^{3+} to interfere with pathogen metabolism, with their toxicity primarily arising from non-specific disruption of iron metabolic pathways. Ga^{3+} can competitively bind to iron-binding proteins such as siderophores and transferrin, thereby hindering normal iron uptake and transport. This leads to iron deficiency *in vivo*, affecting hemoglobin synthesis and the activity of iron-dependent enzymes. Consequently, long-term administration may precipitate iron-deficiency anemia [73].

2.4.2. Monitoring framework for metal ion toxicity

Monitoring metal ion toxicity in peptide-metal chelates requires integrating both *in vitro* prescreening and *in vivo* dynamic monitoring across multiple dimensions, including detection of metal ion release and accumulation, cytotoxicity assessment, monitoring of organ function, and evaluation of molecular-level damage.

Detection of metal ion release kinetics can be performed using techniques such as inductively coupled plasma mass spectrometry (ICP-MS) and atomic absorption spectroscopy (AAS) to quantitatively measure the amount and rate of metal ion release from chelates under various conditions, including simulated physiological environments (pH7.4 PBS), tumor microenvironments (pH6.5–6.8), intracellular lysosomal environments (pH4.5–5.0), and in serum. These assessments provide critical insights into coordination stability and the risk of metal ion liberation [74]. For the specific detection of the redox states of copper ions, techniques such as X-ray absorption near-edge structure (XANES) and chemical sensors can be employed to achieve *in situ* imaging and speciation analysis of Cu^+ and Cu^{2+} [75]. Speciation analysis of radionuclides can be performed using high-performance liquid chromatography coupled with inductively coupled plasma tandem mass spectrometry (HPLC-ICP-TQ-MS) to investigate their oxidative transformation and protein corona formation in cell culture media [76].

In vivo accumulation studies are conducted using inductively coupled plasma mass spectrometry (ICP-MS) to quantitatively determine metal ion concentrations in blood and major organs, including the brain, kidneys, liver, spleen, and bones, at various time points post-administration [77]. Laser ablation-inductively coupled plasma mass spectrometry (LA-ICP-MS) enables spatial distribution imaging of metal ions in tissue sections. When coupled with cryo-electron microscopy and metal-ion fluorescent probes, it facilitates *in situ* localization of metal ions at subcellular resolution [78].

Cytotoxicity screening can be performed using CCK-8 or MTT assays to evaluate the inhibition rate of chelates on normal cell proliferation and determine the half-maximal inhibitory concentration (IC₅₀). For DNA-damaging metal ions such as copper and platinum, the Comet assay is employed to assess the extent of DNA strand breaks [79]. To assess cuproptosis-related mechanisms, assays are required to detect lipoylated protein aggregation and the activity of Fe—S cluster proteins. For metal ions that interfere with enzyme activity, changes in the activity of key enzymes and proteins, including superoxide dismutase (SOD), glutathione peroxidase (GSH-Px), ceruloplasmin, and transferrin should be measured [80].

Biochemical assessment of organ function includes the measurement of serum alanine aminotransferase (ALT) and aspartate aminotransferase (AST) to evaluate liver function; serum creatinine (Scr), blood urea nitrogen (BUN), and urine microalbumin to assess renal function; complete blood count (CBC) to evaluate bone marrow hematopoietic function; and cardiac troponin and lactate dehydrogenase (LDH) to assess myocardial function [81], [82], [83]. Histopathological examination involves paraffin embedding, sectioning, and hematoxylin and eosin (HE) staining of major accumulation organs to observe histomorphological changes. Immunohistochemistry (IHC) is employed to detect the expression of apoptosis-related proteins (Caspase-3, Bax/Bcl-2), oxidative stress-related proteins (Nrf2, HO-1), and DNA damage markers (γ-H2AX) [84].

2.4.3. Mitigation strategies for metal ion toxicity

Enhancing the coordination stability of peptide-metal chelates represents a fundamental strategy to reduce free metal ions and mitigate toxicity. This approach begins at the molecular design stage by incorporating multidentate coordinating amino acids, such as histidine, cysteine, and aspartic acid, into the peptide sequence. This strategy constructs high-affinity metal-binding domains and optimizes coordination geometry, thereby improving the overall coordination stability of the chelate [85]. Replacing linear peptides with cyclic or stapled peptides introduces conformational constraints that further enhance structural stability, while N-terminal acetylation and C-terminal amidation modifications effectively improve resistance to enzymatic degradation [86].

At the process regulation level, chelates can be encapsulated within platforms such as pH-responsive metal-organic frameworks (MOFs), GSH-responsive liposomes, or enzyme-responsive self-assembling nanoparticles to achieve controlled release at target sites, thereby reducing exposure of normal tissues [87]. Surface PEGylation reduces immunogenicity and prolongs circulation time, while the introduction of hydrophilic moieties such as polyethylene glycol (PEG) chains or glycosyl groups facilitates renal clearance and minimizes *in vivo* accumulation [88]. Furthermore, incorporating antioxidant amino acids such as tyrosine and tryptophan into the peptide sequence, or covalently attaching antioxidants such as vitamin E and curcumin, enables *in situ* neutralization of reactive oxygen species (ROS), thereby further alleviating oxidative stress damage [89].

At the clinical intervention level, local injection or topical administration can be employed for localized diseases to minimize systemic absorption. The minimum effective dose and maximum tolerated dose should be established based on pharmacokinetic studies, and intermittent dosing regimens may be adopted to provide a time window for the *in vivo* clearance of metal ions [90]. For patients with hepatic or renal insufficiency, dosage adjustments or avoidance of specific chelates are necessary. Concurrent use of antidotes is also recommended, such as glutathione, penicillamine, or EDTA for copper overload; granulocyte colony-stimulating factor (G-CSF) for platinum-induced myelosuppression; hydration, diuresis, and amifostine for platinum-induced

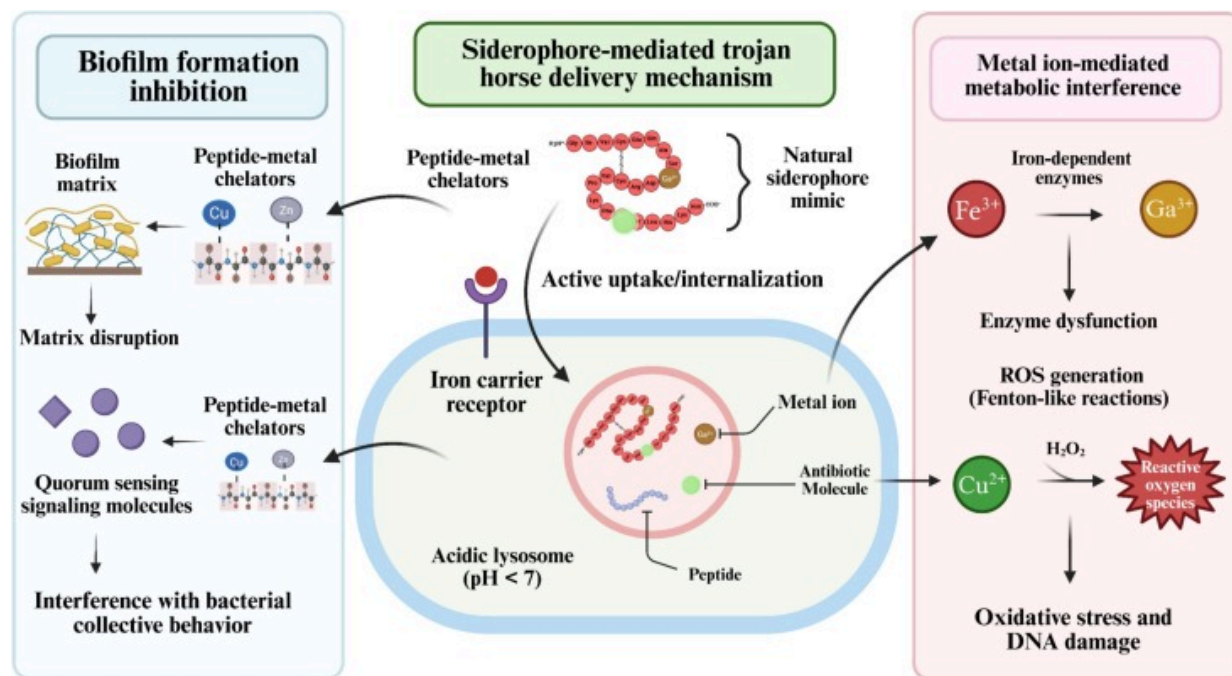
nephrotoxicity; and amifostine or vitamins C and E as radioprotectants for radiation-induced damage.

3. Research on the application of antimicrobial therapy

3.1. Research on antibacterial mechanisms

Instead of relying on a single target, the antibacterial mechanism of peptide-metal chelates establishes a multi-level, multi-dimensional synergistic network that integrates “precise delivery,” “metabolic interference,” and “collective defense disruption.” Confronted with the increasingly formidable physiological defense barriers posed by pathogens, these chelates ingeniously exploit the nutrient acquisition and trace-element metabolic pathways essential for bacterial survival. As comprehensively illustrated in [Fig. 2](#), this tripartite mechanism initiates at the cellular boundary through a siderophore-mediated Trojan horse strategy. The network first breaches the bacterial outer envelope by mimicking natural trace-metal carriers to engage specific iron-carrier receptors. This interaction facilitates the active, targeted internalization of the complex. Upon entering the acidic intracellular lysosomal environment ($\text{pH} < 7$), the complex dissociates, releasing free metal ions, targeted peptides, and potential antibiotic molecules. Following this internalization, profound metal ion-mediated metabolic interference takes effect. The released specific metal ions fundamentally disrupt core bacterial metabolic pathways and redox homeostasis. For instance, gallium ions (Ga^{3+}) competitively displace iron (Fe^{3+}) to cause iron-dependent enzyme dysfunction, while copper ions (Cu^{2+}) catalyze Fenton-like reactions to generate abundant reactive oxygen species (ROS), inflicting lethal endogenous oxidative stress and DNA damage. Simultaneously, the mechanism operates entirely extracellularly to inhibit biofilm formation. By interacting with the external environment, the peptide-metal chelators intercept quorum sensing signaling molecules and disrupt the protective biofilm matrix, thereby depriving bacteria of the physical and biological foundations necessary for collective behaviors. This all-encompassing attack strategy, spanning extracellular to intracellular environments and single cells to

multicellular populations, has demonstrated remarkable bactericidal efficacy and substantial potential to overcome antibiotic resistance in theoretical studies and *in vitro* models.



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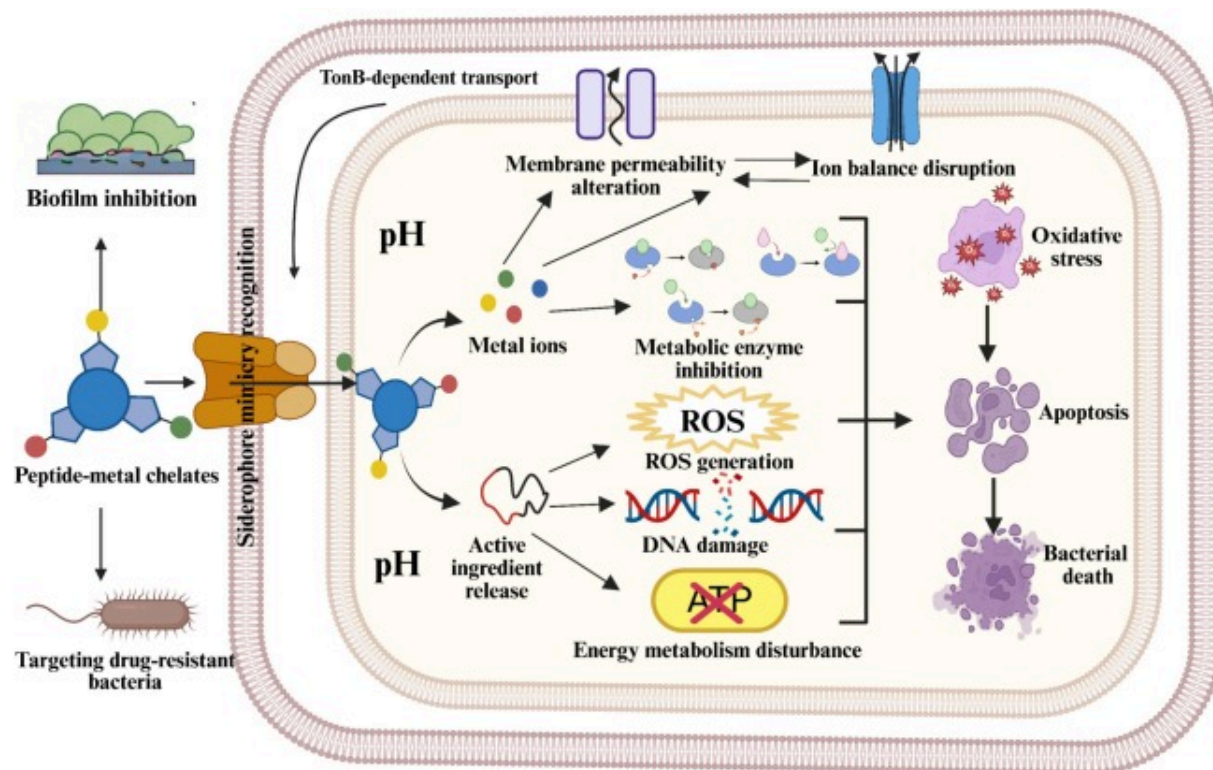
Fig. 2. Schematic illustration of the tripartite synergistic antibacterial mechanism of peptide-metal chelates.

3.1.1. Siderophore-mediated Trojan horse delivery mechanism

The antibacterial mechanism of peptide-metal chelates involves a complex, multi-level, multi-target interaction network, with iron carrier-mediated bacterial iron uptake systems constituting the most central pathway. By mimicking the molecular structure and coordination properties of natural bacterial iron carriers, these chelates can be

actively recognized and taken up by bacteria, thereby enabling the specific delivery of antibacterial components [91].

The iron-carrier-mediated Trojan horse strategy represents the most significant antibacterial mechanism of such chelators. As shown in Fig. 3, bacteria have evolved highly specific iron uptake systems to acquire iron, an element essential for growth yet extremely low in bioavailability. By precisely mimicking the three-dimensional structure and metal coordination geometry of natural iron carriers, peptide-metal complexes can “deceive” bacterial iron carrier receptors, leading them to be mistaken for nutrients and actively internalized [48]. Once inside the bacterial cell, the chelate structure dissociates in the acidic lysosomal environment, releasing antibacterial metal ions or coupled antibiotic molecules to achieve precise bacterial killing. Research indicates that iron-carrier- β -lactam conjugates significantly enhance antibiotic activity against Gram-negative bacteria through this mechanism [92].



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Fig. 3. Mechanistic insights into the antimicrobial action of peptide-metal chelates.

3.1.2. Metal ion-mediated metabolic interference synergistic mechanism

Disruption of metal ion metabolism represents another critical mechanism. Gallium ions (Ga^{3+}), owing to their ionic radius and coordination properties similar to those of iron ions (Fe^{3+}), can substitute for iron in bacterial metabolic processes. However, their lack of redox activity impairs the function of iron-dependent enzyme systems [32]. Copper ions (Cu^{2+}) generate reactive oxygen species (ROS) through Fenton-like reactions, inducing oxidative stress and DNA damage [30].

3.1.3. Auxiliary mechanism for biofilm formation inhibition

The mechanism of biofilm formation inhibition involves disrupting bacterial quorum sensing and extracellular matrix synthesis [93]. The mechanism of biofilm formation inhibition involves disrupting bacterial quorum sensing and extracellular matrix synthesis. Peptide-metal chelators can affect bacterial adhesion and biofilm structural stability by chelating divalent metal ions such as Ca^{2+} and Mg^{2+} [94]. At the same time, certain metal complexes can interfere with the synthesis or recognition of quorum-sensing signaling molecules, thereby inhibiting bacterial collective behavior.

Although the aforementioned mechanisms are highly compelling in theory and have been validated in simplified models, the challenges they face in real, complex infection environments cannot be overlooked. First, bacterial populations exhibit formidable evolutionary adaptability. Prolonged selective pressure may rapidly induce resistance to these novel mechanisms, such as through mutations in iron transporter receptors, upregulation of efflux pumps, or expression of competitive metal-chelating molecules. Second, the biofilm environment common in chronic infections, characterized by internal heterogeneity and dormant bacteria, may severely undermine strategies reliant on active uptake or metabolic activity. Furthermore, most *in vitro* studies fail to adequately account for the complex interactions between the host immune system and metal ions/peptides, which can substantially influence the distribution, stability, and ultimate efficacy of chelating agents. Consequently, the evaluation of these mechanisms must extend beyond idealized *in vitro* conditions and be conducted within complex physiological and pathological settings that more closely approximate clinical realities.

3.1.4. Evolutionary counter-strategies and resistance mitigation design

While the “Trojan horse” strategy offers a promising route to bypass traditional resistance mechanisms, its long-term clinical viability hinges on anticipating and countering bacterial evolutionary responses. Empirical evidence confirms that pathogens can rapidly evolve resistance to siderophore-mediated delivery systems through multiple molecular pathways. The most prominent mechanism involves receptor mutation and

downregulation. The primary entry points for therapeutic conjugates, such as specific siderophore receptors like *P. aeruginosa* FptA or CirA, are genetically mutable. Clinical isolates resistant to cefiderocol have been shown to harbor loss-of-function mutations in these receptor genes, which directly abrogate the uptake of siderophore-antibiotic conjugates [95]. In some cases, regulatory mutations lead to global downregulation of siderophore receptor expression. This downregulation confers cross-resistance even in the absence of structural receptor changes [96]. Therefore, structural and regulatory adaptations at the receptor level constitute a primary defense barrier for bacteria.

Beyond receptor modifications, bacteria employ competitive siderophore secretion as an adaptive ecological defense. Pathogens may secrete high-affinity endogenous or interspecies siderophores, such as pyoverdine, that outcompete therapeutic conjugates for receptor binding. Notably, bacterial populations evolved under drug pressure can increase their natural siderophore production *via* specific regulatory shifts. This overproduction displaces the therapeutic agent from its target receptors and shields susceptible kin in the community, a phenomenon termed cooperative cross-protection [97]. Consequently, this collective behavioral adaptation significantly limits the long-term efficacy of relying solely on a single transport pathway.

Furthermore, bacteria can neutralize internalized therapies by upregulating efflux pumps and intracellular detoxification mechanisms. Upon internalization, microbes may activate broad-spectrum efflux systems to expel the conjugate before drug release. Additionally, metal-specific resistance can arise from the expression of metallothioneins or periplasmic chelators that sequester liberated therapeutic metal ions [98]. This active sequestration blunts the metabolic disruption intended by the treatment. Such multifaceted internal defense systems underscore the need to design therapies that can systematically evade or overwhelm intracellular clearance mechanisms.

To proactively mitigate these evolutionary trajectories, researchers are developing rational design strategies that address multiple resistance pathways simultaneously, with combination therapy emerging as a highly effective approach. Co-administration of a

Trojan horse conjugate with an agent targeting a non-overlapping vulnerability, such as a cell wall synthesis inhibitor, significantly narrows the mutant selection window. This combination makes simultaneous resistance evolution statistically improbable [99]. For instance, testing cefiderocol combined with aztreonam has demonstrated remarkable synergy against carbapenem-resistant strains. This dual approach delayed the emergence of resistance in serial passage experiments, demonstrating the value of targeting multiple distinct cellular targets [100].

Another sophisticated design strategy involves the development of multi-stimuli responsive systems. Engineering conjugates that require two or more independent stimuli for activation, such as acidic pH coupled with elevated glutathione and bacterial-specific enzyme cleavage, drastically reduces the probability of spontaneous resistance. A recent example is a dual-responsive gallium-peptide nanoconjugate activated only by both bacterial esterase for deprotection and intracellular low pH for dissociation. This specific formulation showed no detectable resistance after multiple generations in bacterial cultures, unlike generic single-stimulus controls [101]. These intelligent trigger mechanisms ensure that the therapeutic payload is released only in the intended microenvironment, reducing opportunities for the bacteria to adapt.

Finally, introducing target redundancy *via* receptor promiscuity provides a critical blueprint for sustained drug uptake. Designing conjugates that bind multiple distinct siderophore receptors ensures continued internalization even if one specific receptor is mutated. Moreover, metallo-sideromycins incorporating bismuth citrate not only enhance primary efficacy but also suppress resistance development in *P. aeruginosa* by modulating membrane permeability and oxidative stress [102]. This illustrates the profound power of metallodrug synergism in overcoming bacterial defenses. Ultimately, integrating these anti-evolutionary design principles is essential for translating peptide-metal chelates into durable therapeutics. Future work must prioritize evolutionary forecasting methods that use experimental evolution platforms combined with deep sequencing to comprehensively map resistance landscapes before widespread clinical deployment.

3.2. Representative antibacterial peptide-metal chelate complexes

Peptide-metal chelates have evolved into multiple representative structural types in the field of antimicrobial therapy, each featuring unique construction strategies and application characteristics. These representative systems not only demonstrate the diversity in peptide-metal chelate design but also offer novel solutions for clinical anti-infective treatment.

Iron-carrier-antibiotic conjugates represent the most maturely researched class of compounds. Cefiderocol, as the first approved iron-carrier-cephalosporin conjugate, signifies a breakthrough in this technology. Cefiderocol achieves active drug transport by leveraging bacterial iron uptake systems through the covalent linkage of the cephalosporin β -lactam ring to a catechol-type iron carrier [103]. This design enables cefiderocol to exhibit superior activity against multidrug-resistant Gram-negative bacteria, including carbapenem-resistant strains. It enters bacterial cells *via* TonB-dependent outer membrane transporters, bypassing outer membrane permeability barriers and efflux pumps. Several orders of magnitude can reduce its minimum inhibitory concentration (MIC) values compared to traditional antibiotics [104].

Metal-polyphenol framework antibacterial systems leverage the reductive properties and strong coordination abilities of polyphenolic compounds to synergistically construct multifunctional materials with regular pore structures through coordination with metal ions. These systems typically form through the coordination self-assembly of natural polyphenols, such as gallic acid, tannic acid, or dopamine, with metal ions such as iron, copper, or zinc [105], [106]. Metal-polyphenol frameworks not only exhibit excellent biocompatibility and degradability but also enable spatiotemporal control of antimicrobial activity through pH-responsive release of metal ions and polyphenol molecules [107]. Research indicates that such systems can significantly enhance the removal efficiency of mature biofilms and effectively inhibit bacterial adhesion and aggregation [108].

Multifunctional metal-complex nanoplatforms integrate targeting, therapeutic delivery, and imaging capabilities within a single system. Typically centered on metal nanoparticles or metal-organic frameworks (MOFs), these platforms feature surface modifications with targeting peptides, cell-penetrating peptides, or antibody fragments to achieve precise recognition of specific pathogens [109]. Through rational design, these nanoplatforms can generate thermotherapy, sonodynamic, or magnetothermal effects under external stimuli such as near-infrared laser, ultrasound, or magnetic fields, synergistically enhancing antimicrobial efficacy when combined with chemotherapy.

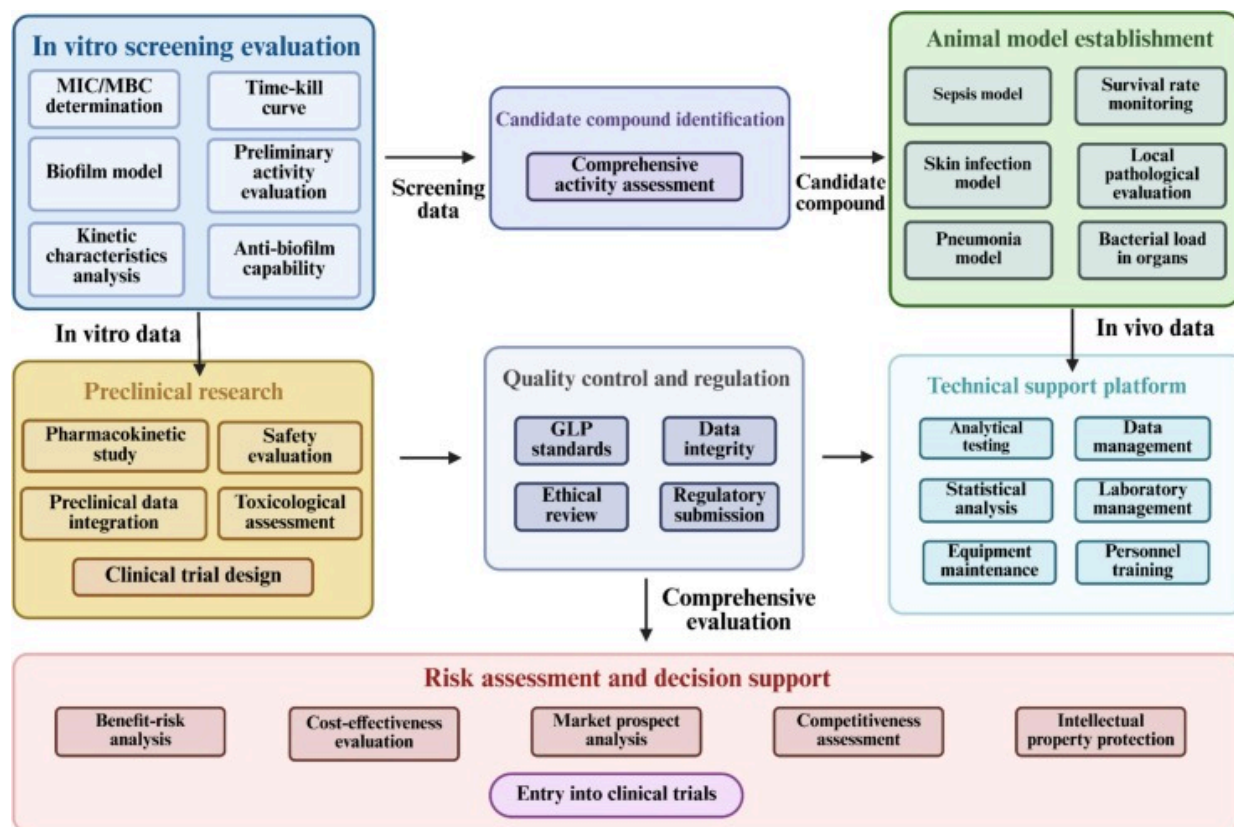
Smart-responsive antimicrobial materials can automatically modulate drug release behavior in response to changes in the infection microenvironment. Such materials are typically engineered with chemical bonds or structural units that are sensitive to stimuli such as pH, enzymes, redox potential, or temperature [110]. Under normal physiological conditions, the drug remains stable. Still, its release is triggered by a specific acidic environment, the presence of specific enzymes, or high concentrations of reactive oxygen species at the infection site. This intelligent response significantly enhances treatment precision and safety while reducing systemic toxic side effects.

The successful development of these representative systems marks a significant shift in peptide-metal chelate research from fundamental studies to practical applications. Future trends will increasingly focus on multifunctional integration and intelligent control, achieving synergistic effects by rational combinations of functional modules to provide more innovative solutions to address the growing challenge of antimicrobial resistance.

3.3. Evaluation of antibacterial effects *in vitro* and *in vivo*

The evaluation of antimicrobial efficacy for peptide-metal chelates requires establishing a comprehensive assessment system that covers the entire process, from *in vitro* screening to *in vivo* validation. As illustrated in Fig. 4, the evaluation of antimicrobial efficacy for peptide-metal chelates follows a systematic workflow progressing from *in vitro* screening to *in vivo* validation, ultimately guiding clinical translation. This process begins with

preliminary *in vitro* activity screening, in which compound activity is initially assessed using measures such as minimum inhibitory concentration (MIC). This is complemented by kinetic characterization using time-kill curves and evaluation of biofilm eradication capacity. Upon identifying candidate compounds, research advances to *in vivo* efficacy evaluation. This involves establishing multiple animal models, such as mouse sepsis, pneumonia, and skin infection models, to comprehensively monitor survival rates, bacterial loads in organs, and histopathological changes, and thereby fully validate *in vivo* antimicrobial efficacy and safety. Subsequently, preclinical studies will further integrate pharmacokinetic and systemic toxicological assessments to provide critical evidence for clinical dose design and administration regimens of candidate compounds. This system not only delivers key data for compound optimization but also establishes a foundation for safety evaluation and dose design in preclinical research.



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Fig. 4. A comprehensive framework for evaluating antimicrobial efficacy.

Minimum inhibitory concentration (MIC) testing is a fundamental indicator of antimicrobial activity, conducted using standardized broth or agar dilution methods. The conventional broth dilution method employs serial twofold dilutions in test tubes to determine the lowest drug concentration that inhibits bacterial growth. In contrast, the microbroth dilution method uses 96-well plates for high-throughput screening, significantly enhancing experimental efficiency [111]. Agar dilution can simultaneously test multiple strains, making it particularly suitable for sensitivity screening of clinical isolates [112]. All methods must adhere to CLSI or EUCAST standard guidelines, employing

quality control strains such as *Escherichia coli* ATCC 25922 and *Staphylococcus aureus* ATCC 29213 to ensure data reliability and comparability.

The study of bactericidal kinetics reveals the time-dependent characteristics of drug action through time-kill curves. This method involves incubating a specific concentration of an antimicrobial agent with the test bacterial suspension, performing colony counts at 0, 2, 4, 6, 8, 12, and 24h, and plotting a logarithmic curve of time *versus* colony count [113]. The slope of the curve reflects the rate of bactericidal action, while the area under the curve (AUC) quantifies the overall bactericidal efficacy. Synergy is defined as a reduction in bacterial count of $\geq 2 \log_{10}$ CFU/mL when drugs are combined compared to monotherapy, whereas antagonism is defined as an increase of $\geq 2 \log_{10}$ CFU/mL. This dynamic evaluation method distinguishes between bacteriostatic and bactericidal agents and provides crucial guidance for designing dosing regimens.

Animal infection model validation is a critical component of preclinical research. By establishing mouse models of sepsis, pneumonia, or skin infection, the *in vivo* efficacy and safety of drugs can be evaluated. The tail vein injection method is used to establish a sepsis model, with drug efficacy assessed by monitoring survival rates, organ bacterial load, and levels of inflammatory cytokines [114]. Intratracheal inoculation establishes a pneumonia model to evaluate bacterial clearance and lung pathology [115]. These models can simulate the pathological processes of human infection, providing important references for predicting clinical dosages and optimizing administration regimens.

The evaluation of biofilm removal efficacy requires the use of specific biofilm models to assess a drug's ability to penetrate and eradicate mature biofilms. The integrated Pdots sensor array based on fluorescence resonance energy transfer (FRET) can be combined with machine learning to identify and characterize bacterial biofilms efficiently, providing a novel technical means for accurate biofilm detection and rapid evaluation of their eradication efficacy [116]. The Calgary biofilm apparatus, or 96-well plate static model, is commonly used for biofilm cultivation. Biofilm quantity is quantified *via* crystal violet staining, or bacterial viability within the biofilm is observed through live/dead staining

[117]. Evaluation metrics include minimum biofilm eradication concentration (MBEC), biofilm inhibition rate, and eradication rate. For chronic infection treatment, biofilm assessment is critically important because bacteria within biofilms can exhibit reduced antimicrobial susceptibility by 10 to 1000 times compared to planktonic bacteria [118].

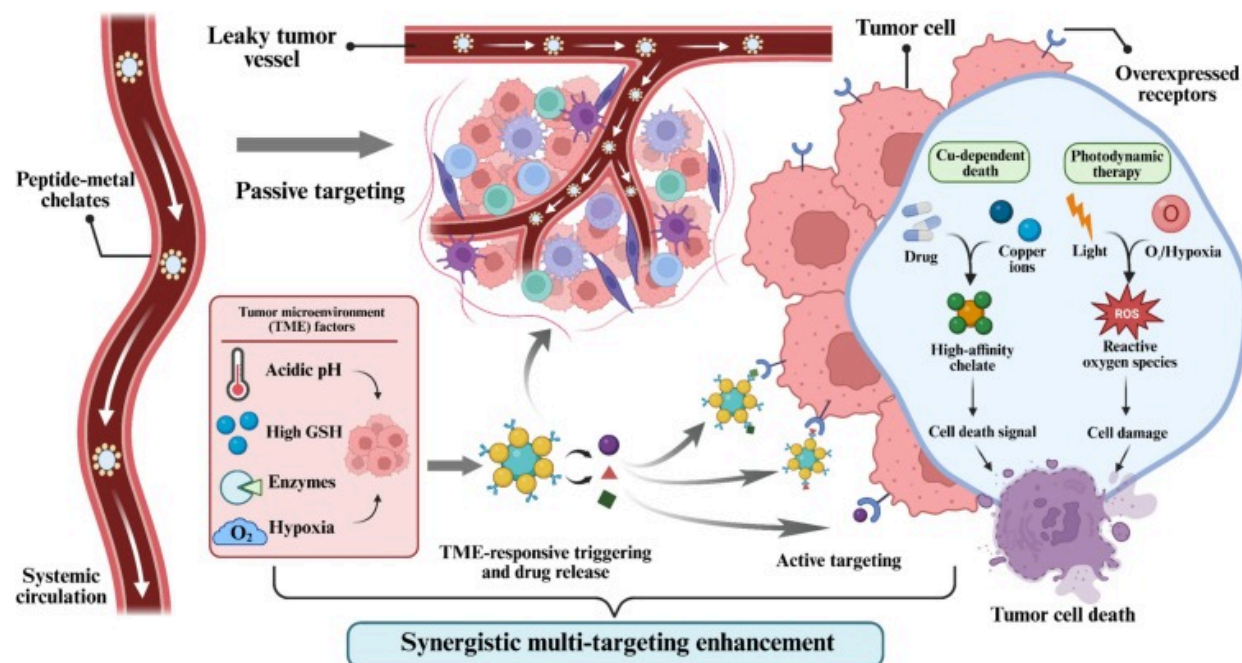
By integrating *in vitro* and *in vivo* evaluation data with pharmacokinetic parameters and safety metrics, a comprehensive assessment of the clinical value of peptide-metal chelates can be achieved. This multi-level, multi-indicator evaluation system not only ensures the scientific rigor and reliability of the data but also provides a solid foundation for subsequent clinical translation. The ongoing advancement of novel technologies and methodologies, including microfluidic chips, organ-on-a-chip systems, and artificial intelligence-driven predictive models, is poised to significantly enhance the precision and efficiency of antimicrobial efficacy evaluation. This progress promises to provide substantial impetus for the development of new anti-infective therapeutic agents.

4. Applied research in tumor treatment

4.1. Tumor targeting mechanism

The successful application of peptide-metal chelates in oncological therapy hinges fundamentally on their ability to accumulate selectively in malignant tissues while minimizing systemic toxicity. To navigate the formidable physiological barriers presented by solid tumors, researchers have constructed a sophisticated multi-tiered targeting framework that capitalizes on the distinct pathophysiological hallmarks of cancer. As comprehensively delineated in Fig. 5, this integrated therapeutic journey begins within the systemic circulation. Peptide-metal chelates first exploit the leaky architecture of tumor blood vessels to achieve initial passive accumulation *via* the enhanced permeability and retention effect. Upon extravasation into the tumor interstitium, these complexes encounter a unique set of biochemical factors characterizing the tumor microenvironment, including localized acidic pH, elevated glutathione levels, highly expressed specific enzymes, and hypoxia. These localized stimuli serve as intelligent

switches, triggering responsive structural transformations and precise payload release. Subsequently, the system engages in active targeting; the exposed functional ligands bind specifically to overexpressed receptors on the tumor cell membrane, facilitating efficient cellular internalization. Within the intracellular compartment, parallel customized cytotoxic mechanisms are unleashed to maximize cell death. For example, specific precursor drugs can selectively bind transition metals, such as copper ions, to form high-affinity chelates, thereby initiating potent copper-dependent cell death. Concurrently, responsive systems can exploit localized conditions combined with external stimuli, such as light, to activate photodynamic therapy pathways and generate lethal reactive oxygen species. The spatiotemporal integration of passive enrichment, microenvironment-responsive triggering, and receptor-mediated internalization culminates in a profound synergistic multi-targeting enhancement. Nevertheless, the profound heterogeneity of human tumors and the intricate realities of the *in vivo* microenvironment severely complicate the clinical translation of such static sequential targeting designs. Consequently, the field is undergoing a critical paradigm shift toward the development of intelligent adaptive systems.



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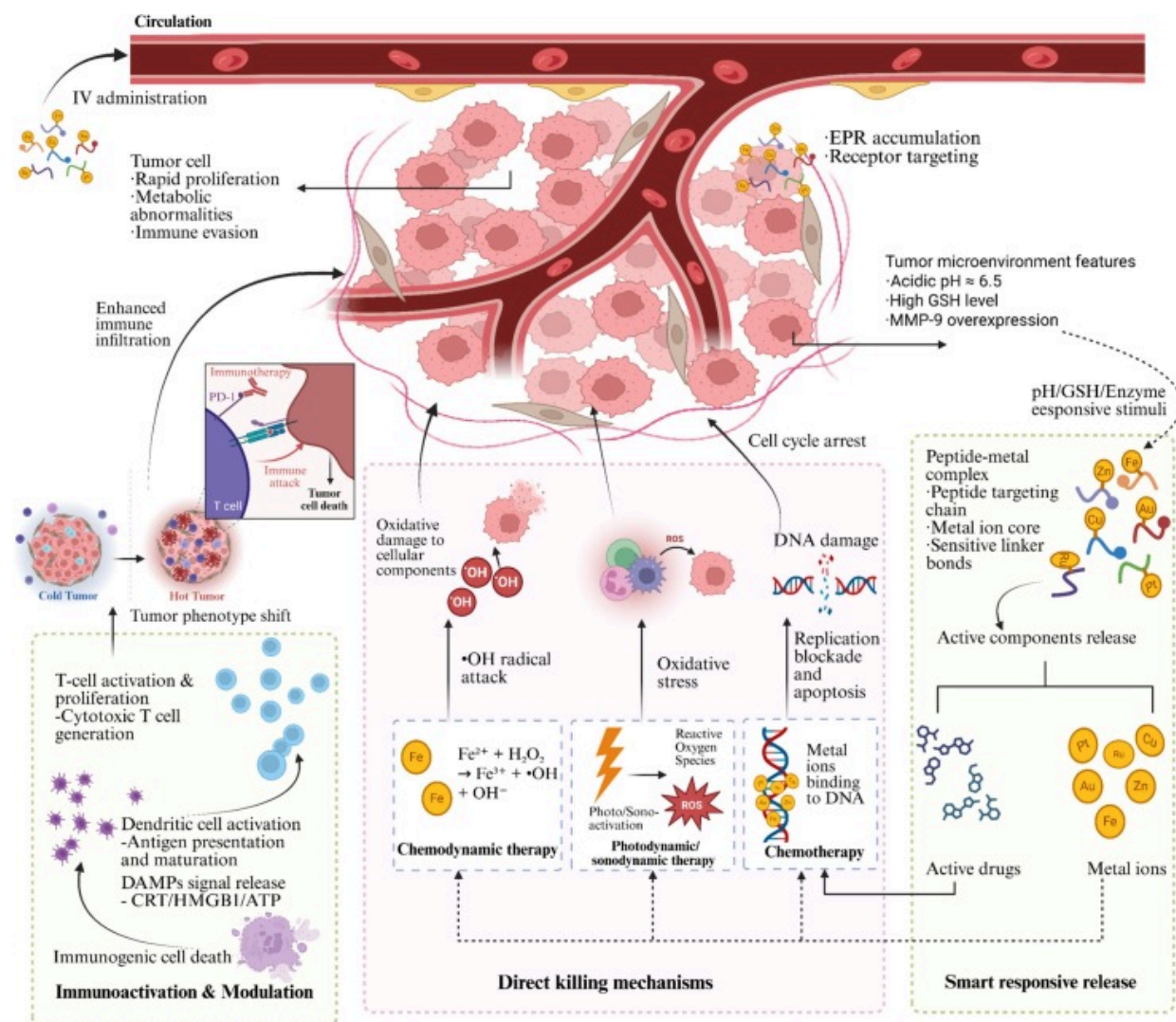
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Fig. 5. Synergistic multi-targeting enhancement mechanisms of peptide-metal chelates in tumor therapy.

4.1.1. Passive targeting of tumor enrichment mediated by the EPR effect

The targeted mechanism of peptide-metal chelates in tumor therapy primarily relies on the unique physiological characteristics of the tumor microenvironment, achieving selective drug accumulation in tumor tissues through passive and active targeting strategies, as illustrated in Fig. 6. The enhanced permeability and retention (EPR) effect refers to the phenomenon wherein macromolecules or nanoparticles selectively accumulate in tumor tissues due to the leaky vasculature and impaired lymphatic drainage of solid tumors [119]. It serves as the core mechanism of passive targeting, stemming from the anatomical abnormalities of the tumor vasculature [120]. Tumor

tissues exhibit extensive angiogenesis with high vascular density but irregular morphology, lacking a smooth muscle layer and a functional lymphatic system. This leads to prolonged retention of extravasated macromolecular drugs. Sluggish venous return further enhances the accumulation effect of these drugs [121].



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Fig. 6. Multimodal antitumor mechanisms of peptide-metal chelates.

4.1.2. Precision recognition triggered by the tumor microenvironment

The chemical properties of the tumor microenvironment provide ideal conditions for active targeting. The acidic microenvironment of tumor tissue (pH6.5–6.8) creates conditions conducive to pH-responsive drug release [122]. High concentrations of glutathione (GSH) trigger redox-responsive systems [123]. Receptor-mediated endocytosis enables active targeting by integrating receptor ligands overexpressed on tumor cells. For instance, folate receptors, which are highly expressed on the surface of various tumor cells, serve as specific targets [124].

Active targeting strategies also include enzyme-responsive systems that trigger drug release by tumor-specific enzymes such as matrix metalloproteinases MMP-2 and MMP-9. For instance, a study targeting triple-negative breast cancer engineered MMP-2-responsive nanodroplets that undergo PEG decoating upon enzymatic degradation in the tumor microenvironment, significantly enhancing nanoparticle penetration and accumulation within tumor tissues. The therapeutic core of this system lies in the active metabolite of the encapsulated drug, disulfiram, which forms high-affinity chelates with copper ions, thereby inducing copper-dependent death in tumor cells [125]. Additionally, the dual mechanism of integrin receptor-mediated targeting and tumor-penetrating peptides, such as iRGD, can increase drug accumulation by ≥ 2 -fold [126].

4.1.3. Synergistic enhancement of multi-targeting strategies

The synergistic application of multi-targeted strategies significantly enhances therapeutic efficacy. Research indicates that nanoplatforms integrating EPR effects with active targeting achieve tumor accumulation levels exceeding those of non-targeted systems by more than twofold [127]. The intelligent, responsive system dynamically modulates drug release behavior in response to multiple stimuli within the tumor microenvironment (hypoxia, acidic pH, elevated GSH), achieving a paradigm shift from oxygen-dependent Type II photodynamic therapy to hypoxia-tolerant Type I photodynamic therapy [128].

A deep understanding and synergistic optimization of these targeting mechanisms lay a solid foundation for the application of peptide-metal chelates in tumor therapy, providing crucial technical support for precision medicine. Tumor-targeting strategies based on the EPR effect and active targeting are the cornerstone of peptide-metal chelate design. Yet the clinical universality of this foundation is increasingly under scrutiny. Human tumors exhibit extreme heterogeneity, with the EPR effect showing significant variability across patients, tumor types, and even different regions within the same tumor. This leads to average tumor delivery efficiencies of nanomedicines in clinical settings often falling below 1%, far below those reported in animal models. The baseline expression of receptors used for active targeting (such as integrins and folate receptors) in normal tissues may also lead to off-target uptake, resulting in toxicity. Furthermore, the triggering conditions for tumor microenvironment-responsive release (e.g., pH- or enzyme-responsive) may not be as “precise” or “clear-cut” *in vivo* as simulated *in vitro*, posing risks of incomplete release or premature leakage. Consequently, future targeting strategies must shift from reliance on single, static targets toward “intelligent adaptive” systems capable of adapting to the dynamic, heterogeneous tumor ecosystem, complemented by more precise patient stratification.

4.1.4. Engineering approaches to clinical translation challenges

Although multi-stage targeting strategies have established an elegant theoretical blueprint for drug delivery, their clinical translation confronts formidable challenges posed by tumor heterogeneity, dense stroma, and low delivery efficiency. The extreme heterogeneity of human tumors results in significant inter-patient and inter-tumoral variability in the enhanced permeability and retention (EPR) effect, even across different regions within the same tumor. Clinical data indicate that the average tumor delivery efficiency of nanomedicines often falls below 1% of the administered dose [129]. Furthermore, the dense tumor extracellular matrix (ECM) forms a physical barrier that severely impedes the deep penetration of nanoparticles into tumor tissues [130]. In response to these bottlenecks, the research focus in this field is shifting from static “multi-stage design” to dynamic “engineering strategies” that aim to overcome *in vivo*

penetration and distribution barriers by precisely regulating chelate size, surface properties, and delivery pathways.

Precise regulation of dynamic, reversible changes at the nanoscale is a core strategy for overcoming penetration barriers. Conventional nanoparticle size design requires a trade-off between prolonged circulation and deep tumor penetration; however, the heterogeneity of vascular endothelial pores in tumors makes it difficult for a single size to achieve both objectives simultaneously [131]. To address this, researchers have developed nanosystems that can be size-transformed. For instance, nanoparticles maintain a relatively large hydrodynamic diameter during systemic circulation to reduce renal clearance. However, upon reaching the tumor microenvironment, they rapidly disassemble or shrink into ultra-small nanoparticles triggered by locally elevated matrix metalloproteinases (MMP-2/MMP-9) or acidic pH, thereby significantly enhancing their diffusion capacity within the dense interstitial matrix [132]. This “from large to small” smart, responsive strategy introduces a new design dimension that enables peptide-metal chelates to penetrate deep into the core regions of tumors and exert their therapeutic effects.

Optimizing surface physicochemical properties is critical for enhancing tumor accumulation and penetration. Surface charge and hydrophilicity are two core determinants of the *in vivo* behavior of nanoparticles. Although conventional PEGylation effectively prolongs circulation time, it may hinder cellular uptake due to the so-called “PEG dilemma” [133]. Studies have shown that introducing topologically embedded functional groups into partially PEGylated nanocarriers can enhance tumor accumulation by more than twofold without altering the overall PEG density. This improvement is attributed to the fine regulation of the nanoparticle surface hydration layer and protein corona composition by the embedded moieties [134]. Furthermore, modulation of surface topography has also been demonstrated as an effective strategy to enhance tumor uptake. Compared to conventional nanoparticles with smooth surfaces, those with rough surface topography, self-assembled from bottlebrush block copolymers, exhibit significantly

prolonged circulation time and enhanced tumor accumulation and distribution due to their unique hydrodynamic behavior [135].

In response to the uncertainties surrounding the EPR effect in human tumors, a novel concept, “active transport and retention” (ATR), has been proposed. Studies indicate that nanoparticle entry into tumors involves not only passive diffusion through endothelial gaps but also active transcytosis across tumor endothelial cells and transport *via* tumor-associated macrophages [136]. Based on these findings, engineering design can alternatively leverage specific biological pathways, including screening and targeting “nanoparticle-transporting endothelial cells” that highly express transport-related genes, as well as genetic engineering of normal endothelial cells to confer transport capability. These approaches open new avenues for nanomedicine delivery that do not depend solely on the EPR effect [127].

The dense extracellular matrix (ECM) composed of collagen and hyaluronic acid in the tumor microenvironment represents a key physical barrier that impedes drug penetration. To overcome this barrier, researchers have developed various prodrug strategies that modulate ECM components *in situ* to remodel the tumor matrix. For instance, co-delivery of anti-fibrotic agents such as losartan with peptide-metal chelates can inhibit ECM synthesis, thereby reducing solid tumor pressure, collapsing blood vessels, improving tumor perfusion, and ultimately enhancing the delivery efficiency and distribution uniformity of nanomedicines [137]. Furthermore, targeting cancer-associated fibroblasts (CAFs) to deplete or reverse their function is an effective strategy to suppress ECM production and open channels for drug penetration [138].

In summary, given the practical obstacles in clinical translation, the design of peptide-metal chelates must abandon reliance on single, static targeting and adopt an “intelligent adaptive” system that integrates multiple engineering approaches, such as dynamic size transformation, surface property optimization, active transcellular transport, and microenvironment regulation. The synergistic application of these strategies is expected

to fundamentally overcome the penetration and distribution barriers to tumors and elevate targeting efficiency from its current bottleneck to a level with clinical significance.

4.2. Therapeutic modalities and mechanisms

Peptide-metal chelates exhibit diverse therapeutic modes in cancer treatment, each with unique mechanisms of action and application characteristics. The chemotherapeutic mode directly induces tumor cell apoptosis by leveraging the inherent cytotoxicity of metal complexes. For instance, a study on half-sandwich iridium(III) complexes demonstrated that structural modifications of the benzimidazole ligand can precisely modulate their biological activity. Among them, a neutral iridium(III) complex designated as 1[C, NH-Cl] specifically targets and accumulates in mitochondria. It induces mitochondrial proton leak, decreases membrane potential, and increases reactive oxygen species (ROS) levels, ultimately triggering regulated cell death. [139]. Copper-based complexes generate reactive oxygen species *via* Fenton-like reactions, triggering oxidative stress and lipid peroxidation, ultimately inducing ferroptosis [140].

Photodynamic therapy (PDT) is a therapeutic modality based on photosensitizing metal complexes that generate reactive oxygen species upon exposure to specific wavelengths of light. Iridium(III) complexes serve as ideal photosensitizers due to their exceptional photophysical properties. They promote intersystem crossing *via* spin-orbit coupling, prolonging triplet lifetimes and enhancing the singlet oxygen quantum yield [141]. Research indicates that heterometallic Ru(II)-Ir(III) binuclear photosensitizers can achieve dual Type I and Type II photosensitization effects. Under normal oxygen conditions, they yield singlet oxygen at a rate of 0.76–0.78 mol/mol, while still generating hydroxyl radicals and superoxide anions in hypoxic environments, effectively overcoming tumor hypoxia limitations [142].

Chemokinesis-directed therapy (CDT) leverages Fenton or Fenton-like reactions to convert hydrogen peroxide overexpressed in tumors into highly toxic hydroxyl radicals. Copper-based complexes sustain $\cdot\text{OH}$ generation through $\text{Cu}^{2+}/\text{Cu}^{+}$ redox cycling, inducing

tumor cell death [30]. The smart responsive system leverages the acidic pH and high GSH levels in the tumor microenvironment to achieve controlled release of metal ions and regulate reaction rates, thereby enhancing treatment specificity and reducing side effects.

Acoustic-driven therapy (ADT) utilizes ultrasound to activate photosensitizers, overcoming the limitations of light penetration depth. Research indicates that the photosensitization properties of metal complexes can be effectively regulated by designing organic ligands. For instance, thiophene-based ionic iridium(III) complexes exhibit significantly extended triple excited state lifetimes due to heavy atom effects and ligand modifications. This enables efficient generation of reactive oxygen species under ultrasonic irradiation, demonstrating superior DPBF oxidation degradation rates compared to conventional photosensitizers and outstanding potential for sonodynamic therapy [143]. SDT generates ROS through multiple mechanisms, including acoustic cavitation, sonoluminescence, and mechanical microfluidics, making it particularly suitable for deep-seated tumor treatment.

Immunomodulatory therapy enhances antitumor immune responses by regulating immune cells through metal ions. Iridium(III) complexes induce immunogenic cell death (ICD), releasing damage-associated molecular patterns (DAMPs), including calreticulin, ATP, and MCP-1, thereby activating dendritic cells and cytotoxic T lymphocytes [144]. Copper-mediated cell death has been confirmed as a potent method for inducing an immune-mediated killing response, capable of triggering a sustained immune reaction and transforming “cold tumors” into “hot tumors” [145]. The tumor-specific nanoimmunomodulator, constructed using a cascade activation strategy, can mediate phototherapy to achieve tumor eradication and induce immunogenic cell death upon irradiation with an 808nm near-infrared laser. It simultaneously triggers the *in-situ* release of the caged Toll-like receptor 7/8 agonist, which is further activated by intracellular NQO1 to release the bioactive agonist. Through the cascade effect of phototherapy and immunomodulation, the nanoimmunomodulator significantly potentiates systemic immune activation and enhances the infiltration efficiency of cytotoxic T lymphocytes and helper T cells, thereby providing a novel engineering

paradigm for the precise implementation of immunomodulatory therapy [146]. As key pattern recognition receptors, Toll-like receptors 7/8 (TLR-7/8) have agonists that are highly promising immunomodulatory agents for tumor immunotherapy, capable of effectively activating the innate immune response. Yet uncontrolled administration often induces severe immune-related adverse events (irAEs). Nanodrug strategies serve as a core solution for the precise delivery and tumor microenvironment-specific activation of TLR-7/8 agonists, and their mechanistic research and clinical translation have further enriched target selection and strategic design in immunomodulatory therapy [147].

Beyond inducing immunogenic cell death, directly reshaping the intrinsically immunosuppressive tumor microenvironment (TME) by eradicating the lactic acid-shaped acidic niche represents a paramount strategy for next-generation immunomodulators. For instance, an innovative multichannel immune nanoregulator engineered with calcium peroxide (CaO_2) nanoreactors and liposome coatings has been developed to specifically address this challenge [148]. By integrating ultrasound-triggered oxygen bursts and endogenous H^+ -responsive CaO_2 degradation, this system dynamically depletes local protons and inhibits glycolysis-associated lactic acid production. This targeted modulation effectively cuts off the source of acidity, thereby polarizing immunosuppressive M2 macrophages toward a pro-inflammatory phenotype and restoring the activity of cytotoxic T lymphocytes. Such an advanced strategy of uprooting the root causes of TME immunosuppression perfectly complements the delivery of immunomodulators, highlighting a profound avenue for metal-involved nanoplatforms in comprehensive solid tumor therapy.

Combination therapy strategies achieve synergistic effects and reduced toxicity by coordinating the actions of different mechanisms. Photodynamic-immunotherapy combination treatment utilizes PDT to induce programmed cell death (ICD) while simultaneously alleviating immune suppression, thereby enhancing the efficacy of PD-1/PD-L1 inhibitors [149]. Chemokine immunotherapy reshapes the tumor microenvironment by synergistically inducing potent immune responses *via* ferroptosis and cupric-mediated cell death [150].

Furthermore, the application of nanotechnology to modulate the immunosuppressive tumor microenvironment and enhance T lymphocyte infiltration extends beyond traditional immune checkpoint combinations; it is currently revolutionizing engineered cellular immunotherapies. For instance, recent advances in nano-/nanobio-technology have been heavily leveraged to sequentially and spatially reprogram CAR-T cell therapies, successfully addressing overarching bottlenecks in solid tumors, such as off-target toxicity and the immunosuppressive barrier [151]. By exploiting synthetic genetic circuits, bio-orthogonal chemical switches, and biomimetic tissue scaffolds, these nano-engineering strategies enable immune cells to dynamically sense and adapt to the tumor ecosystem, thereby shifting the therapeutic paradigm toward a spatiotemporally controlled, multimodal treatment regimen.

The advancement of nanoplatform technology provides a fundamental carrier for integrating therapeutic modalities. Metal-organic frameworks (MOFs) as nanocarriers, featuring tunable pore structures and high specific surface areas, enable efficient drug encapsulation and controlled release [152]. The liposome-peptide conjugation system combines the biocompatibility of liposomes with the targeting properties of peptides, thereby enhancing drug delivery efficiency and therapeutic efficacy [153]. The application of these advanced delivery systems enables more precise and efficient therapeutic approaches for peptide-metal chelates.

4.3. Representative antitumor peptide-metal chelate complexes

In the field of antitumor therapy, peptide-metal chelates have evolved into diverse and well-defined structural systems, each designed with specific targeting mechanisms and therapeutic modalities. As summarized in Table 3, these systems can be broadly categorized based on their metal centers and functional mechanisms. Representative examples include endoplasmic reticulum-targeting Ir(III) complexes, mitochondria-localized two-photon photosensitizers, and pH-activated prodrugs, each demonstrating distinct design rationales and promising antitumor efficacy.

Table 3. Representative antitumor peptide-metal chelate systems.

Representative compound	Metal center	Features	Therapeutic application	Ref.
Endoplasmic reticulum-targeted iridium complex	Ir3	Iridium(III)	Targets endoplasmic reticulum, induces type I/II photodynamic effects and pyroptosis, and activates antitumor immunity.	[144]
Heterobimetallic photosensitizer	Ru(II)-Ir(III)	Ruthenium(II)/Iridium(III)	Heterobimetallic type I/II dual-effect photosensitizer that produces multiple ROS under normoxic/hypoxic conditions and overcomes tumor hypoxia.	[142]
Mitochondria-targeted iridium-based oncosis inducer	Ir-swelling inducer	Iridium(III)	Targets mitochondria, induces caspase-independent immunogenic oncosis, overcomes multidrug resistance and activates cGAS-STING pathway.	[154]
Iridium complex inducing ferroptosis-like death	Ir-dF	Iridium(III)	Photoactivation initiates proton-coupled electron transfer, induces iron- and oxygen-independent lipid peroxidation, and triggers ferroptosis-like death.	[155]
Mitochondria-targeted two-photon iridium complex	Ir4	Iridium(III)	Under hypoxia, reduced two-photon excitation generates carbon radicals that damage mitochondria, inducing apoptosis.	[156]

Representative compound	Metal center	Features	Therapeutic application	Ref.
Nucleus-targeted iridium complex	Polypyridine Iridium(III) Complex	Iridium(III)	Penetrates the cell nucleus, causes DNA damage under light irradiation, and arrests the cell cycle.	[157]
pH-activated iridium prodrug	Ir-pH-prodrug	Iridium(III)	Activated in the tumor's mildly acidic microenvironment, releases active components, achieving selective killing.	[158]
Iridium(III)-based photothermal agent	Ir-NPs	Iridium(III)	High photothermal conversion efficiency, near-infrared light excitation generates heat and ablates tumors; it can synergistically deliver drugs.	[159], [160]
Ruthenium(II) polypyridyl complex	TLD-1433	Ruthenium(II)	Phase II clinical photosensitizer with high singlet-oxygen yield targets bladder cancer.	[161]
Ruthenium-Platinum heterobimetallic complex	Ru-Pt	Ruthenium(II)/Platinum(II)	Integrates photodynamic therapy with DNA crosslinking, overcomes cisplatin resistance, and synergistically kills.	[162], [163]
Photocatalytic hydrogen-producing ruthenium system	Ru-cat@NPs	Ruthenium(II)	Photocatalytically decomposes water to produce hydrogen within tumors, acts as an antioxidant, alleviates hypoxia, and sensitizes therapy.	[164]

Representative compound	Metal center	Features	Therapeutic application	Ref.
Rhenium(I) carbonyl complex	Re-CO	Rhenium(I)	Releases CO gas molecules, is anti-inflammatory, promotes apoptosis, and also has phosphorescence imaging function.	[165]

Functionalized nanoplatforms based on ruthenium complexes exhibit excellent biocompatibility and diverse therapeutic functions. For instance, researchers have designed heterometallic Ru(II)-Ir(III) binuclear photosensitizers that retain the same ligand framework. By adjusting the metal centers, the properties of the photosensitizers can be effectively modulated, achieving highly efficient tumor cell killing under both normoxic and hypoxic conditions [142]. These platforms not only deliver superior photodynamic therapy effects but also modulate the tumor microenvironment through metal coordination, thereby enhancing immune responses.

Copper-based complex diagnostic-therapeutic integrated systems combine therapeutic and imaging functions within a single platform. The IrP-L@V2O5 nanosystem developed by Cheng Li et al. [166] simultaneously conjugates the microenvironment-modulating losartan and the hypoxia-responsive fluorescent iridium complex probes to vanadium pentoxide nanoparticles. By synergistically combining *in situ* oxygen delivery with microenvironment regulation, it addresses tumor hypoxia during staged photodynamic therapy. This platform also enables real-time monitoring of tumor oxygen levels during treatment, providing therapeutic guidance.

In parallel with the development of metal-based optical and magnetic imaging systems, advanced luminescent materials are also being profoundly explored to optimize *in vivo* imaging quality. For instance, epoxy-type afterglow luminescent probes, powered by a cascade reaction of photochemical defects *via* epoxy intermediates, exhibit a continuous luminescence phenomenon after excitation. This unique mechanism affords extremely

impressive signal-to-background ratios (SBR) *in vivo*, rendering them uniquely suited for biological disease diagnosis and theranostics [167]. The design principles and outstanding predictive analysis capabilities of such high-SBR afterglow materials offer valuable inspiration for integrating ultrasensitive optical tracking modalities into future peptide-metal theranostic platforms.

Smart responsive materials can automatically modulate their functions based on disease states. For instance, studies indicate that the endoplasmic reticulum-targeted iridium(III) complex IrC can induce immunogenic cell death in melanoma cells through the endoplasmic reticulum stress pathway. This process releases damage-associated molecular patterns, activates dendritic cells and cytotoxic T lymphocytes, and significantly enhances the efficacy of PD-1 monoclonal antibody immunotherapy. This establishes a research foundation for metal-based drug immunotherapy [168].

Bacterial vector-mediated delivery systems leverage the tumor-targeting properties of microorganisms to achieve precise drug delivery. A research team developed the iridium photosensitizer-bacterial hybrid (Ir-HEcN) by genetically engineering *Escherichia coli* Nissle 1917 to overexpress HaloTag protein on its surface. The iridium complex forms a covalent bond with the HaloTag protein *via* a chlorinated hexane structure, achieving stable anchoring. Leveraging bacterial tumor colonization properties, this hybrid exhibits superior tumor targeting and penetration. Under 525nm light irradiation, it induces tumor cell pyroptosis and promotes immune cell infiltration [169].

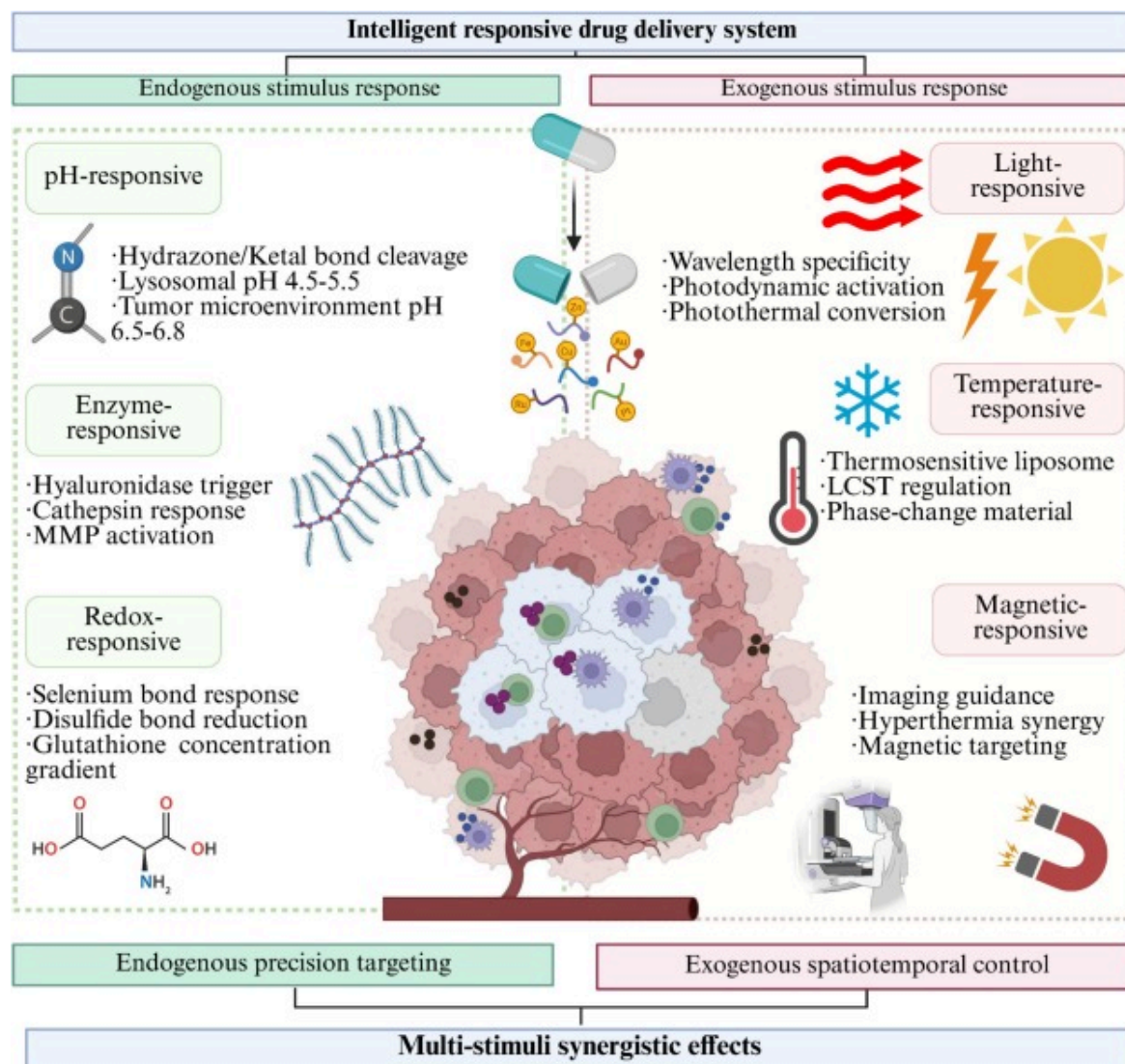
The successful development of these representative systems marks a significant shift in peptide-metal chelate research from fundamental studies to practical applications. Notably, the emergence of multifunctional integrated platforms has achieved synergistic effects through rational combinations of functional modules, offering innovative solutions to address challenges posed by tumor heterogeneity and therapeutic resistance. Future development trends will increasingly focus on the construction of integrated diagnostic and therapeutic platforms and on the development of personalized treatment strategies.

By combining smart responsive materials with precision delivery technologies, these efforts aim to realize true precision medicine.

5. Drug delivery system

5.1. Intelligent responsive delivery system

Smart responsive drug delivery systems represent one of the core innovative directions in peptide-metal chelate research. These systems enable precise, controlled drug release by sensing specific pathophysiological changes in the microenvironment of diseased sites [170]. Such delivery systems are primarily constructed based on two types of response mechanisms, as illustrated in Fig. 7. One category involves endogenous stimulus-responsive mechanisms, encompassing physiological and pathological signals inherent to the disease microenvironment, such as pH levels, redox potential [171], and enzyme activity [172]. The other category comprises exogenous stimulus-responsive mechanisms, involving externally controllable triggers such as light exposure, magnetic fields [173], and temperature. These mechanisms significantly enhance the targeting and safety of drug therapy.



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Fig. 7. Design of stimuli-responsive drug delivery systems based on peptide-metal chelates.

The core design principle of pH-responsive drug delivery systems is to leverage the pH gradient between the tumor microenvironment and intracellular acidic compartments, as well as between these compartments and normal tissues, to achieve targeted, controlled drug release. *In vitro* and *in vivo* studies confirm that the pH of the extracellular tumor microenvironment typically ranges around 6.8. At the same time, intracellular acidic compartments such as lysosomes and endosomes exhibit pH values as low as 5.0 [174], [175]. Based on this, a study developed a mild hyperthermia platform centered on Ti₃C₂ MXene nanosheets, combining it with quercetin (Qu), which possesses heat shock protein 70 (HSP70) inhibitory activity. This nanoscale system exhibits precise pH-responsive drug release, rapidly releasing the drug in an acidic environment at pH5.0. The cumulative release rate reached 91.34% within 48h, thereby effectively enhancing tumor cell sensitivity to photothermal therapy [176]. Separate research has successfully synthesized novel two-dimensional IL-Ti₃C₂T_x MXene nanosheets. This material exhibits strong light absorption in the near-infrared region, achieving a photothermal conversion efficiency of 63.91%, making it a highly promising photothermal therapeutic agent. Additionally, its remarkable pH sensitivity enables efficient drug release in acidic tumor microenvironments, thereby further enhancing antitumor therapeutic efficacy [177].

Enzyme-responsive drug delivery systems trigger drug release by leveraging enzymes specifically overexpressed in tumors or bacteria, significantly enhancing therapeutic targeting specificity. The high expression of hyaluronidase (HAase) in the tumor microenvironment enables an intelligent delivery system using hyaluronic acid-modified mesoporous silica nanoparticles. This design actively targets CD44-overexpressing cancer cells through its HA shell, which HAase subsequently degrades to trigger precise drug release. Experimentally, this system induced potent early and late apoptosis in 43% and 55% of cells, respectively, and demonstrated antitumor efficacy threefold higher than the free drug and twofold higher than non-targeted nanoparticles, validating its responsive and targeted therapeutic advantage [178]. The mPEG-GFLG-DOX micelle, which responds to cathepsin B, removes its PEG shell through specific peptidic bond cleavage by the enzyme, thereby promoting uptake by tumor cells. The encapsulated drug synergizes

with the physically encapsulated drug, significantly enhancing cytotoxicity [179]. Additionally, esterase-responsive lipid nanocarriers release PD-1 inhibitors upon the action of tumor-overexpressed esterases, achieving 24-fold higher bioavailability than conventional formulations and enabling targeted, synergistic anticancer effects [180].

The redox-responsive controlled-release system exploits the disparity between high intracellular glutathione (GSH) concentrations in tumor cells and the extracellular environment. Intracellular GSH levels (2–10mM) exceed extracellular concentrations (2–20μM) by over 200-fold, providing a specific triggering condition for disulfide bonds and other reduction-sensitive chemical bonds [181]. GSH-responsive copper-disulfiram nanoparticles can be reduced by GSH within tumor cells, generating highly toxic complexes *in situ* and releasing the drug, thereby significantly enhancing chemotherapy efficacy [182]. The GGT/GSH dual-responsive PLGA-SS-PEG-Glu nanoplatfrom enhances penetration into the tumor microenvironment *via* enzyme-mediated charge reversal, thereby promoting GGT-induced charge switching and GSH-responsive GA release. [183], The GSH-responsive PROTAC nanocapsule MPL(SS)P@SIS rapidly releases drugs *via* disulfide bond cleavage, degrades BRD4, depletes GSH to elevate ROS levels, and synergistically prolongs survival in glioblastoma mouse models when combined with chemoradiotherapy [184].

The multi-stimulus-responsive system integrates multiple triggering mechanisms to achieve precise, controlled drug release and synergistic enhancement of antitumor therapy. The pH/GSH dual-responsive CuFeMOF@GOx@Mem bimetallic nanocatalytic system releases Fe³⁺ and Cu²⁺ under acidic conditions in the tumor microenvironment. Further activated by intracellular high-concentration GSH, it enhances chemotherapeutic efficacy through three mechanisms, including self-sustained H₂O₂ production, catalyst self-recycling, and GSH consumption. This system achieves over 90% *in vitro* tumor cell-killing efficiency. [185]. pH/GSH/NIR Triple-Responsive CHO-SMNPs Magnetic Nanocarriers Featuring dual-sensing design *via* imine and disulfide bonds, these carriers exhibit outstanding photothermal conversion efficiency under 808nm near-infrared irradiation. In simulated tumor conditions (pH5.5, 10mM GSH), drug release rates are

threefold higher than in physiological conditions, enabling synergistic antitumor effects through combined chemotherapy and photothermal therapy [173].

Smart responsive systems represent an advanced direction in drug delivery technology, but their “smartness” currently remains largely confined to responding to a single or a few predefined stimuli. The *in vivo* environment is a complex network teeming with diverse biomolecules, ions, and physical signals. These exogenous or endogenous interference signals may cause non-specific triggering or signal masking, compromising release precision. For instance, the acidification level of the tumor microenvironment fluctuates dynamically and overlaps with inflammatory regions; glutathione (GSH) levels also exhibit variability between individuals and across tissues. Furthermore, many response mechanisms, especially those that rely on bond cleavage, are irreversible once activated. This irreversibility limits their utility in applications that demand precise control over drug dosage. Consequently, the development of next-generation smart systems equipped with advanced features such as multi-signal logic gating, adaptive feedback regulation, or reversible response capabilities represents a crucial direction toward achieving truly precise drug delivery.

5.2. Mechanistic insights into dynamic coordination

The discussion above highlights how structural features of peptide-metal chelates dictate their responsiveness to microenvironmental stimuli. However, a deeper mechanistic understanding requires moving beyond static descriptions to probe the dynamic behavior of metal coordination under biologically relevant conditions. Recent advances in computational chemistry and time-resolved spectroscopy have begun to unravel these dynamics, revealing that coordination geometries fluctuate on timescales relevant to biological processes and that these fluctuations directly impact metal selectivity, reactivity, and ultimately therapeutic efficacy [186].

Computational approaches have proven particularly powerful in predicting how coordination geometries evolve in response to environmental changes. Long-time

molecular dynamics simulations combined with quantum calculations have demonstrated that the conformational entropy of the peptide backbone plays a critical role in modulating metal-binding affinity. For instance, studies on aluminium binding to short disordered peptides revealed that significant entropic penalties accompany metal coordination, and that strategies such as proline incorporation or sequence length optimization can effectively tune this entropic cost [187]. Integrated approaches combining docking, quantum mechanics, and molecular dynamics have further enabled prediction of how non-covalent interactions within the peptide scaffold influence the metal coordination sphere, with single-point mutations shown to trigger reversible ligand-induced conformational changes analogous to those observed in metalloproteins [188]. These computational predictions have been validated through free energy calculations that demonstrate the spontaneity of peptide-metal complex formation and its dependence on coordination geometry [189].

Time-resolved spectroscopic techniques have provided direct experimental evidence for the existence of transient coordination intermediates that are inaccessible to conventional structural methods. Stopped-flow spectroscopy studies on Cu^{2+} binding to ATCUN (amino-terminal copper and nickel) motifs revealed that complex formation proceeds *via* partially coordinated 2N intermediates that exist within time windows of hundreds of milliseconds to seconds [186]. Crucially, these intermediates exhibit fundamentally different reactivity compared to the final inert 4N square-planar complex: they readily undergo metal exchange with competing ligands such as histidine. They are rapidly reduced by physiological reductants, such as glutathione [186]. This finding uncovers a previously unrecognized layer of kinetic control over metal bioavailability and reactivity, with direct implications for both physiological metal handling and the design of therapeutic chelators.

The concept of kinetic inertia, in which thermodynamic stability does not guarantee kinetic inertness, has emerged as a critical consideration for predicting the fate of metal-peptide complexes in complex biological environments [190]. Unlike the static picture provided by equilibrium measurements, these dynamic studies reveal that transient

species may be the true determinants of biological activity, whether in catalytic applications or in the context of metal-induced toxicity. Understanding this dynamic speciation, encompassing both conformational fluctuations and transient intermediate states, is therefore essential for the rational design of next-generation chelators with tunable kinetic properties and predictable *in vivo* behavior [191]. Recent comprehensive reviews have further emphasized that conformational analysis must be integrated into chelating peptide design to achieve desired biological activities [192].

These mechanistic insights directly inform the design of smart responsive delivery systems. The same principles that govern transient intermediate reactivity, including ligand exchange kinetics, redox lability, and conformational dynamics, can be harnessed to engineer stimuli-responsive linkers that release payloads with precise spatiotemporal control in response to specific biological cues.

5.3. Construction of a nanoscale delivery platform

The construction of nanodelivery platforms serves as the core technological foundation for achieving efficient drug delivery *via* peptide-metal chelates. As shown in Table 4, various platforms leverage their precisely regulated structural properties and functionally oriented design advantages to provide diversified solutions for therapeutic needs across different scenarios.

Table 4. Comparative analysis of different nano-delivery platforms.

Platform type	Material composition	Structural characteristics	Drug loading capacity	Advantages
Metal-Organic Frameworks (MOFs)	Metal ions/Organic ligands	Porous crystalline structure	High specific surface area, tunable pore size.	High loading capacity, precisely tunable pore good stability.

Platform type	Material composition	Structural characteristics	Drug loading capacity	Advantages
Liposome Systems	Phospholipids/Cholesterol	Bilayer vesicles	Can load both hydrophilic and hydrophobic drugs.	Excellent biocompatibility, biodegradable, and low immunogenicity.
Self-Assembling Peptides	Functional peptides	Nanofibers/Micelles	Highly designable, multiple loading types.	Biodegradable, non-immunotoxic, custom structure.
Exosome Carriers	Natural vesicles (cell-derived)	Lipid bilayer structure	Efficient loading of biomolecules.	Low immunogenicity, targeting, excellent biocompatibility.
Polymeric Micelles	Amphiphilic polymers	Core-shell structure	Strong hydrophobic drug loading capacity.	Outstanding solubilization, long circulation time, modifiable for targeting.
Inorganic Nanoparticles	Silica/Metals	Mesoporous/Solid structure	Diverse loading types.	High stability, easy functionalization/modification, good imaging properties.

Metal-organic frameworks (MOFs) serve as ideal nanocarriers for peptide loading and targeted delivery due to their tunable pore sizes, high specific surface areas, and flexible functionalization potential. These materials form ordered porous crystalline structures through the coordination self-assembly of metal ions/clusters with organic ligands, enabling precise regulation of drug loading and release behavior. GSH-responsive Zr-based MOF-modified PD-1 inhibitor peptide AUNP12 triggers drug release upon high-concentration GSH exposure in tumors, synergizing with photothermal therapy to activate anti-tumor immunity [193]. Copper-based MOF-199 loaded with miR-4521 releases Cu^{2+} and nucleic acid drugs upon degradation, synergistically inducing tumor cell apoptosis through chemokine therapy and gene therapy [194]. D-mannose-functionalized MgAl-LDH/Fe-MOF nanocomposites co-loaded with methotrexate and doxorubicin target tumors and achieve synergistic dual-drug delivery [195].

The liposome-iron carrier conjugate system combines the biocompatibility of liposomes with the active targeting properties of iron carriers, making it an excellent platform for efficient drug delivery. Liposomes are spherical vesicles formed by phospholipid bilayers, capable of encapsulating both hydrophilic and hydrophobic drugs. Iron carriers, acting as natural or synthetic iron chelators, enable active targeting through iron transport receptors highly expressed on pathogens or tumor cells, significantly enhancing drug accumulation at target sites. In practical applications, iron-carrier-modified liposomes loaded with ceftazidime can penetrate the outer membrane barrier of *Pseudomonas aeruginosa* by exploiting its iron-uptake mechanism. This reduces the minimum inhibitory concentration (MIC) by 50% compared to free drug, significantly enhancing the inhibitory effect against drug-resistant strains [196]. For *Mycobacterium tuberculosis* infection, iron-carrier-functionalized liposomes loaded with bedaquiline enhance drug accumulation within the bacillus via mycobactin-mediated intracellular transport, overcoming the limitation of inadequate intracellular delivery in conventional formulations [197]. In the field of tumor therapy, superparamagnetic iron oxide

nanoliposomes loaded with doxorubicin and modified with iron carriers (pseudomonoside) combined with MUC1 aptamer targeting not only achieve tumor-specific drug release and growth inhibition in colon cancer models but also enable real-time monitoring of treatment progress *via* magnetic resonance imaging (MRI), thereby integrating diagnosis and treatment [198].

Self-assembling peptide nanomaterials spontaneously form ordered nanostructures through noncovalent interactions such as intermolecular hydrogen bonds and π - π stacking. Their core lies in short amino acid sequences with self-assembly potential, combining excellent biocompatibility, low-cost scalable production, and precisely tunable biological activity, making them an optimal platform for targeted drug delivery. The scrambled RGD hexapeptide scrFmoc-GFFRDG rapidly self-assembles into fibrous hydrogels within 10min under physiological conditions, achieving an 8-fold increase in gelation efficiency compared to conventional sequences. It efficiently encapsulates anti-inflammatory drugs and chemotherapeutic agents. At the same time, its constructed microenvironment promotes cell adhesion and regulates cellular activity [199]. The pH-responsive amphiphilic peptide P14 (AAAAFFHHHGRGD) forms nanomicelles through self-assembly, achieving a doxorubicin encapsulation efficiency of 28.02%. Within the acidic tumor microenvironment (pH4.5), it exhibits a 24-h drug release rate of 73.52% and demonstrates significantly superior inhibitory activity against MCF-7 breast cancer cells compared to free drug [200]. The hexahydroxylysine (H₆)-modified peptide nanomaterial H₆-PFC-ICG self-assembles into a core-shell structure. After loading with curcumin, it enables pH-responsive release while simultaneously providing dual-modality imaging capabilities through fluorescence and ¹⁹F magnetic resonance imaging. This platform facilitates real-time monitoring of drug delivery and tumor metastasis [201].

Exosome-mediated drug delivery systems leverage the inherent delivery properties of naturally derived cell vesicles, making them ideal carriers for targeted delivery of biomolecules. Exosomes are actively secreted 30–150nm lipid bilayer nanovesicles containing bioactive substances. With low immunogenicity, high biocompatibility, and natural properties, they efficiently cross biological barriers, making them particularly

suitable for delivering easily degradable biomolecules such as siRNA, mRNA, and proteins. Engineered exosomes, created through genetic modification or surface chemical modification, further enhance targeting specificity and drug loading efficiency, representing a core development direction for next-generation smart targeted delivery systems. Research confirms that miR-145-loaded mesenchymal stem cell-derived exosomes significantly inhibit colorectal cancer cell proliferation and metastasis by downregulating FSCN1 expression [202]. Dendritic cell exosomes loaded with tumor antigens following TLR-3 stimulation can induce potent CD8⁺ T cell responses that suppress melanoma growth [203].

Extending the concept of antigen delivery, rationally engineered cancer nanovaccines represent another highly promising frontier for active immunotherapy. By utilizing programmable nanomaterials to encapsulate tumor-specific antigens, these nanovaccines can precisely target crucial immune cells, such as dendritic cells and T cells, thereby eliciting potent cytotoxic T lymphocyte responses and preventing tumor recurrence [204]. Integrating customizable peptide-metal frameworks into such nanovaccine formulations could further optimize antigen presentation efficiency and provide robust delivery solutions, ultimately helping to overcome the translational hurdles from animal experiments to clinical trials.

The development of these nanoscale delivery platforms fundamentally overcomes the key limitations of traditional drug delivery systems, such as low bioavailability, poor targeting, and significant toxic side effects, thereby providing robust technical support for multifunctional integrated therapies. In addition to nanoscale delivery systems, macroscopic hydrogel scaffolds with dynamically adaptable structures offer crucial design blueprints for complex tissue regeneration and local infection management. A conceptually advanced trilayer hydrogel was recently engineered for scarless urethral repair [210]. By spatially integrating an anti-fouling layer to prevent bacterial adhesion, a rigid water-resistant barrier to avoid occlusion, and a regenerative layer to expedite cell proliferation, this scaffold dynamically adapts to harsh tissue microenvironments to facilitate epithelial recovery and oppose inflammation. Incorporating multifunctional

peptide-metal chelates into microenvironment-adaptable, highly compartmentalized hydrogel architectures could provide a profound paradigm for developing next-generation localized therapies and advanced tissue-engineering platforms. The iterative advancement of such platforms not only accelerates the clinical translation of various formulations, including chemotherapeutic agents, gene therapies, and immunomodulators, but also holds promise for empowering personalized treatment and advancing precision medicine through integrated diagnosis and therapy, thereby pioneering new pathways for treating complex diseases.

6. Integrated theranostics

6.1. Imaging diagnostic function

Peptide-metal chelates demonstrate unique advantages in imaging diagnostics. Their metallic components serve as contrast agents across multiple imaging modalities, while peptide fragments confer targeted recognition capabilities. This synergistic approach enables precise visualization of diagnostics from the molecular to the tissue level [211].

Magnetic resonance imaging contrast agents enhance tissue contrast by utilizing paramagnetic metal ions such as gadolinium and manganese. Gadolinium ions, possessing seven unpaired electrons outside their nucleus, significantly shorten the longitudinal relaxation time (T_1 value) of water protons through electron-proton relaxation coupling, thereby generating high-intensity imaging signals. Traditional clinical gadolinium-based contrast agents (GBCAs) are categorized into linear and macrocyclic types. Macrocyclic gadopentetate dimeglumine exhibits significantly lower gadolinium deposition risk than linear formulations due to its stable metal-ligand coordination structure. Studies confirm that, at equivalent doses, linear GBCAs result in 18 times greater cerebral gadolinium deposition than macrocyclic agents, with deposition levels positively correlated with the number of injections [212]. Peptide-modified metal chelates further optimize targeted imaging performance. For instance, the ZD2 peptide-modified Gd-DOTA chelate specifically recognizes embryonic fibronectin (EDB-FN), which

is highly expressed in high-risk prostate cancer. It accumulates in tumor tissues *via* receptor-ligand-specific binding. In the PC-3 prostate cancer model, MRI signal enhancement in tumor regions significantly outperformed commercial contrast agents in precisely distinguishing malignant lesions from benign hyperplastic tissue. This provides a novel tool for early stratified diagnosis of high-risk prostate cancer [213].

Positron Emission Tomography (PET) probes leverage a synergistic design that combines “peptide-targeted recognition with radiolabeled metal chelation” to enable precise, quantitative detection of lesion metabolic activity. The c-Met-targeting peptide probe (^{68}Ga)Ga-SMIC-1014, derived from the Kringle 3 domain of hepatocyte growth factor, is DOTA-conjugated to ^{68}Ga with radiochemical purity >95%. In the c-Met-overexpressing HCT-116 tumor model, tumor uptake reached $3.56 \pm 0.26\% \text{ID/g}$ within 30 min, with targeting specificity confirmed by antibody blocking assays [214]. For another tumor-specific target, the (^{68}Ga)Ga-DOTA-BP1 probe targeting BCMA couples DOTA-conjugated ^{68}Ga with the BCMA-specific peptide BP1, exhibiting radiochemical purity >98%. achieved a 30-min tumor uptake rate of $4.21 \pm 0.35\% \text{ID/g}$ in the H929 multiple myeloma model, with its targeting specificity also fully validated by antibody blocking assays [215]. Further expanding into clinical applications, the ^{68}Ga -NOTA-WL12 probe modified with the PD-L1-targeting peptide WL12 demonstrated favorable safety in patients with advanced non-small cell lung cancer. One hour post-injection, the tumor-to-lung tissue signal ratio reached 4.45 ± 1.89 , exhibiting significantly superior imaging contrast compared to conventional ^{18}F -FDG probes [216].

Fluorescence imaging leverages the unique photophysical properties of specific metal complexes to provide high-resolution, highly specific tissue and cell imaging tools for biomedical applications. Through synergistic design, peptide-metal chelates further overcome the limitations of traditional fluorescent probes, such as insufficient targeting and significant background interference. The Ir-cMyc probe, constructed by coupling the c-Myc signaling peptide (PAAKRVKLD) with a cyclometallated iridium(III) complex, exhibits deep red light emission. Leveraging the nuclear localization function of the signaling peptide, it achieves precise targeting of cancer cell nuclei. Confocal fluorescence

microscopy validation demonstrates that nuclear fluorescence intensity was significantly higher than cytoplasmic fluorescence intensity, providing an efficient tool for visualizing tumor-associated transcription regulatory factors. [217], Europium (Eu^{3+}) luminescent complexes based on pyrrolidine macrocyclic ligands maintain outstanding two-photon absorption and emission properties after click-chemistry-mediated conjugation with azide-containing cell-penetrating peptides. They enable efficient cytoplasmic imaging in live HeLa cells at low incubation concentrations as low as 50nM, exhibiting significantly superior luminescence quantum yield and photostability compared to conventional organic fluorescent probes. This establishes a foundation for high-resolution imaging of deep tissues [218].

Multimodal imaging integrates the complementary strengths of different imaging technologies to provide comprehensive multidimensional biomedical information for early disease diagnosis, precise lesion localization, and monitoring treatment efficacy. The design of peptide-metal chelates combines peptide-driven target recognition with the versatile functionality of metal chelation, making them ideal central vehicles for multimodal theranostic platforms. The AGGP probe developed for the prostate-specific membrane antigen (PSMA) target in prostate cancer combines ultrafine gold nanoparticles with a multifunctional chelation system. Mediated by PSMA1-specific peptides for targeted delivery, it enables multimodal synergistic MR/CT/near-infrared fluorescence (NIRF) imaging. It not only exhibits excellent renal clearance properties and biocompatibility but also effectively avoids non-specific uptake by the mononuclear phagocyte system. In early-stage prostate cancer models, it precisely delineates lesion boundaries, significantly enhancing the detection rate of micrometastases [219]. Another representative design is the mitochondrial-targeted polypeptide-based multinuclear chelate probe HCy-Lys(DOTA- Gd_3^+)₃. By covalently coupling an orange-red fluorescent dye (HCy) to a trinuclear DOTA- Gd_3^+ chelating unit *via* a customized peptide scaffold, it retains a high longitudinal relaxation rate ($r_1 = 7.65 \text{ mM}^{-1} \cdot \text{s}^{-1}$) to meet T_1 -weighted MRI imaging requirements while enabling real-time dynamic monitoring of live-cell mitochondria through fluorescent signaling. Its water solubility and biocompatibility

further expand the application scenarios of multimodal imaging at the cellular level [220].

The design of peptide-metal chelate imaging probes not only enhances diagnostic accuracy and specificity but also provides crucial evidence for personalized treatment. With the continuous advancement of molecular imaging technology, these smart probes are progressing toward greater precision, efficiency, and clinical translation, laying a solid foundation for precision medicine.

6.2. Therapeutic monitoring and feedback regulation

The core value of the integrated diagnosis and treatment system lies in leveraging the targeted recognition and multifunctional integration properties of peptide-metal chelates to enable real-time monitoring and dynamic feedback regulation of the therapeutic process. By integrating multidimensional information, it provides data support for precision medicine.

Real-time therapeutic monitoring enables dynamic tracking of drug distribution and treatment response through imaging technologies. As an example, the EphA2-targeted bicyclic peptide theranostic system developed by a research team uses the high-affinity bicyclic peptide Pep-34 as a targeting vehicle. By DOTA-conjugating different radionuclides, diagnostic probes, and therapeutic agents, these agents are constructed. During the diagnostic phase, it achieves precise tumor imaging. During the therapeutic phase, ^{177}Lu -labeled Pep-38 delivers targeted radionuclide therapy. Tumor growth inhibition rates reached 83.6% after a single dose and 98.5% after two doses, with no significant systemic toxicity. This approach achieves a synergistic closed-loop system encompassing “diagnosis-therapy-efficacy assessment” [221].

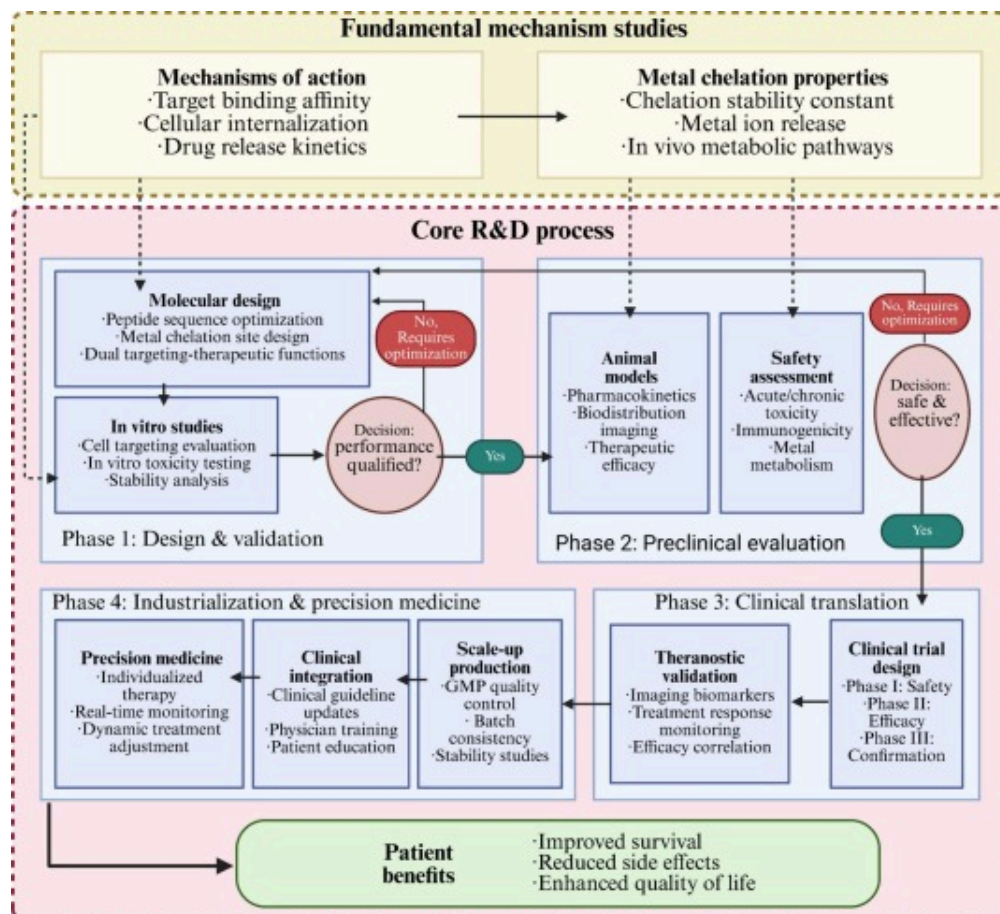
Drug release monitoring and personalized feedback regulation can be achieved through smart responsive probes, such as the D-type peptide PET tracer ^{68}Ga -DOTA-D-P24 developed for the immune checkpoint TIM-3, which enhances *in vivo* stability through D-amino acid modification. It enables quantitative imaging of TIM-3 expression levels

across different tumors. It dynamically reflects real-time changes in target expression during immunotherapy, providing precise molecular imaging evidence for clinicians to adjust treatment regimens [222].

This class of theranostic technology, based on peptide-metal chelates, integrates targeted delivery, imaging monitoring, and feedback regulation to establish a complete precision treatment loop. Through the deep integration of molecular design and imaging techniques, it is pioneering a new paradigm for personalized treatment of malignant tumors and other diseases.

6.3. Representative integrated diagnosis and treatment system

Through innovative engineering, peptide-metal chelate systems seamlessly integrate diagnostic imaging and therapeutic intervention into a single, multifunctional entity. This “theranostic” design principle, which strategically combines the targeting specificity of peptides with the versatile signaling and reactivity of metal ions, is operationalized through a systematic research and development pipeline. As illustrated in Fig. 8, this pipeline encompasses fundamental mechanism studies, core molecular design and preclinical evaluation, and ultimately clinical translation for precision medicine. Guided by this framework, a range of representative technological platforms has been developed. These platforms, ranging from ratiometric probes that provide built-in feedback to stimuli-responsive materials that enable on-demand drug release, are defining the next generation of precision theranostics.



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Fig. 8. Development pathway for theranostic peptide-metal chelate platforms.

Ratio-based theranostic nanoplateforms effectively mitigate background interference from single signals by integrating synchronous detection and quantitative analysis of multiple signaling parameters, significantly enhancing diagnostic accuracy and therapeutic controllability. These platforms utilize peptide-metal chelates as core functional carriers, enabling the seamless integration of targeted recognition, ratio-based imaging, and responsive therapy. The FRET-based ratio-type diagnostic-therapeutic probe P-PpIX

incorporates RGD peptides as targeting units and caspase-3-responsive DEVD peptides as linker arms, integrating FRET fluorophore pairs with the photosensitizer protoporphyrin IX (PpIX). After targeted accumulation in tumor cells overexpressing $\alpha\beta3$ integrin, Photodynamic therapy (PDT)-induced apoptosis activates caspase-3, which cleaves the DEVD peptide, terminating FRET. The ratio of reporter fluorescence (FAM) to reference fluorescence (PpIX) enables precise *in situ* assessment of therapeutic efficacy and caspase-3 expression levels, providing real-time feedback for personalized PDT [223].

Metal-peptide conjugated diagnostic-therapeutic probes achieve organic integration of diagnostic and therapeutic functions through a modular design that combines targeted peptide-mediated delivery and metal-chelation functionalization with therapeutic payload integration, providing an efficient tool for precision medicine as the next-generation precision therapeutic agent following antibody-drug conjugates (ADCs), peptide-drug conjugates (PDCs) offer broad application prospects in the treatment of malignant tumors due to their advantages, including low molecular weight, strong tumor penetration, and controllable production costs. A team developed PDC-2, a PDC targeting integrin $\alpha\beta3$, by conjugating the camptothecin derivative FL118 with the c(RGDfC) peptide. In lung and pancreatic cancer models, it achieved tumor growth inhibition rates of 74.5%–92.3% with a selectivity index exceeding 1000-fold, significantly reducing systemic toxicity [224]. A team's (^{64}Cu)Cu-DOTA-STP stapled peptide probe targeting MDM2/MDMX enables p53 tumor PET imaging and targeted radionuclide therapy *via* DOTA-conjugated ^{64}Cu . It significantly suppresses tumor proliferation, prolongs survival in tumor-bearing mice, and demonstrates favorable biological safety [225]. The clinical-stage ^{68}Ga -HX01 dual-target PET probe simultaneously targets integrin $\alpha\beta3$ and CD13 *via* heterodimeric peptides, enabling clear visualization of lesions in patients with malignant tumors. Tumor SUVmax is significantly higher than that of surrounding tissues, with a low radiation dose and an excellent safety profile [226].

Smart responsive therapeutic materials automatically modulate functional outputs through specific responses to disease microenvironments or exogenous physical signals. Their core advantage lies in integrating multimodal physical energy sources, such as near-

infrared light and ultrasound, to precisely regulate blood-brain barrier permeability, enable targeted drug delivery, and coordinate neural regeneration, thereby providing innovative pathways for treating neurological disorders. Addressing the critical challenge of blood-brain barrier penetration, ruthenium complexes modified with Penetratin peptides and loaded with gold nanostars (Ru@Pen@PEG-AuNSs) demonstrate outstanding performance. This platform enhances BBB penetration efficiency through Penetratin-mediated receptor transport, combines photothermal therapy with A β aggregate disaggregation under near-infrared light irradiation, and enables delivery visualization *via* ruthenium complex fluorescence signaling. In an Alzheimer's disease mouse model, it effectively reduced neurotoxicity and improved cognitive function [227]. In the field of neural stem cell regulation, piezo-nanowired neural stem cells (piezo-NSCs) achieve wireless electrical stimulation by anchoring barium titanate nanowires. Low-intensity ultrasound activation accelerates neuronal differentiation, increasing the proportion of mature neurons from 14.0% to 30.7%. In a rat model of traumatic brain injury, they successfully promote neural circuit reconstruction and restore motor and cognitive functions, providing a remote, precision-control tool for neural repair [228].

The successful development of these representative systems marks a new phase in the advancement of peptide-metal chelate research. As technology continues to mature and clinical data accumulate, integrated diagnostic and therapeutic systems are poised to play a significant role across more disease areas, ultimately driving a paradigm shift from traditional medicine to precision medicine.

6.4. Positioning peptide-metal chelates in the biomedical landscape

The preceding sections have demonstrated the multifunctional capabilities of peptide-metal chelates in imaging, therapy monitoring, and integrated theranostics. However, fully appreciating their unique value requires a systematic comparison with established therapeutic modalities. Table 5 provides a comparative analysis of peptide-metal chelates, antibody-drug conjugates (ADCs), and nucleic acid therapeutics across key parameters relevant to clinical translation.

Table 5. Comparative analysis of peptide-metal chelates, ADCs, and nucleic acid therapeutics.

Parameter	Peptide-Metal Chelates	Antibody-Drug Conjugates (ADCs)	Nucleic Acid Therapeutics
Molecular Weight	1–5kDa (small)	120–160kDa (large)	4–15 kDa (siRNA) to >100kDa (mRNA)
Targeting Precision	High specificity <i>via</i> peptide-receptor recognition	Very high specificity <i>via</i> antibody-antigen binding	Limited; relies on delivery vehicles for targeting
Tumor Penetration	Excellent; small size enables deep tissue diffusion	Poor; limited by size and dense stroma	Moderate; depends on formulation
Blood-Brain Barrier Penetration	Capable; small lipophilic peptides can cross	Incapable; too large to cross intact BBB	Very limited; requires specialized carriers
Immunogenicity	Low to minimal	Moderate to high (anti-drug antibodies)	Low (but innate immune activation possible)
Stability <i>in vivo</i>	Moderate; enhanced by cyclization, D-amino acids	High; engineered for prolonged circulation	Poor; rapidly degraded by nucleases
Manufacturing Complexity	Simple; solid-phase synthesis, scalable	Complex; requires mammalian cell culture, conjugation	Complex; chemical synthesis or enzymatic production
Batch Consistency	High; homogeneous products with defined stoichiometry	Variable; heterogeneous drug-to-antibody ratios	Moderate; depends on synthesis method
Production Cost	Relatively low	Very high	Moderate to high

Parameter	Peptide-Metal Chelates	Antibody-Drug Conjugates (ADCs)	Nucleic Acid Therapeutics
Theranostic Capability	Intrinsic; metal ions provide both therapy and imaging	Limited; requires separate diagnostic counterparts	Minimal; primarily therapeutic only
Clinical Maturity	Emerging; several approved (Lutathera®, PSMA-617)	Mature; >10 approved, blockbuster sales	Emerging; recent approvals for rare diseases
Unique Advantage	Built-in multifunctionality: targeting + metal-based therapy/diagnosis	Potent payload delivery with high specificity	Ability to target “undruggable” genes and pathways
Ref.	[192], [229], [230]	[231], [232]	[233], [234]

This comparative analysis reveals the distinct positioning of peptide-metal chelates within the biomedical landscape. Unlike ADCs, which excel in payload potency but face inherent limitations in tumor penetration and manufacturing complexity due to their large size and heterogeneous nature, peptide-metal chelates offer an optimal balance of molecular size, synthetic accessibility, and targeting specificity [235]. Their low molecular weight enables deep tissue penetration and the ability to cross biological barriers, such as the blood-brain barrier, capabilities that are largely inaccessible to antibody-based therapeutics [236]. Furthermore, the well-established solid-phase peptide synthesis platform ensures batch-to-batch consistency and relatively low production costs compared to the complex mammalian cell culture required for ADCs [237].

Compared to nucleic acid therapeutics, which face challenges of rapid nuclease degradation and require sophisticated delivery vehicles to achieve intracellular delivery, peptide-metal chelates benefit from superior inherent stability and well-characterized formulation strategies [238], [239]. While nucleic acid drugs offer the unique advantage

of targeting previously “undruggable” genes, peptide-metal chelates provide a more direct mechanism of action through metal-mediated cytotoxicity, receptor-targeted radionuclide therapy, or catalytic generation of reactive oxygen species [240].

The most distinctive value proposition of peptide-metal chelates lies in their intrinsic theranostic capacity. Unlike ADCs, which require separate diagnostic counterparts for patient stratification and treatment monitoring, peptide-metal chelates integrate therapeutic and diagnostic functions within a single molecular entity. The metal ion itself can serve as both the therapeutic agent and the imaging reporter, enabling seamless “see and treat” paradigms that are fundamentally difficult to achieve with other modalities [240]. This built-in multifunctionality, combined with synthetic simplicity and favorable pharmacokinetic properties, positions peptide-metal chelates as uniquely versatile tools for precision medicine, particularly in applications requiring real-time therapeutic monitoring and feedback-regulated intervention.

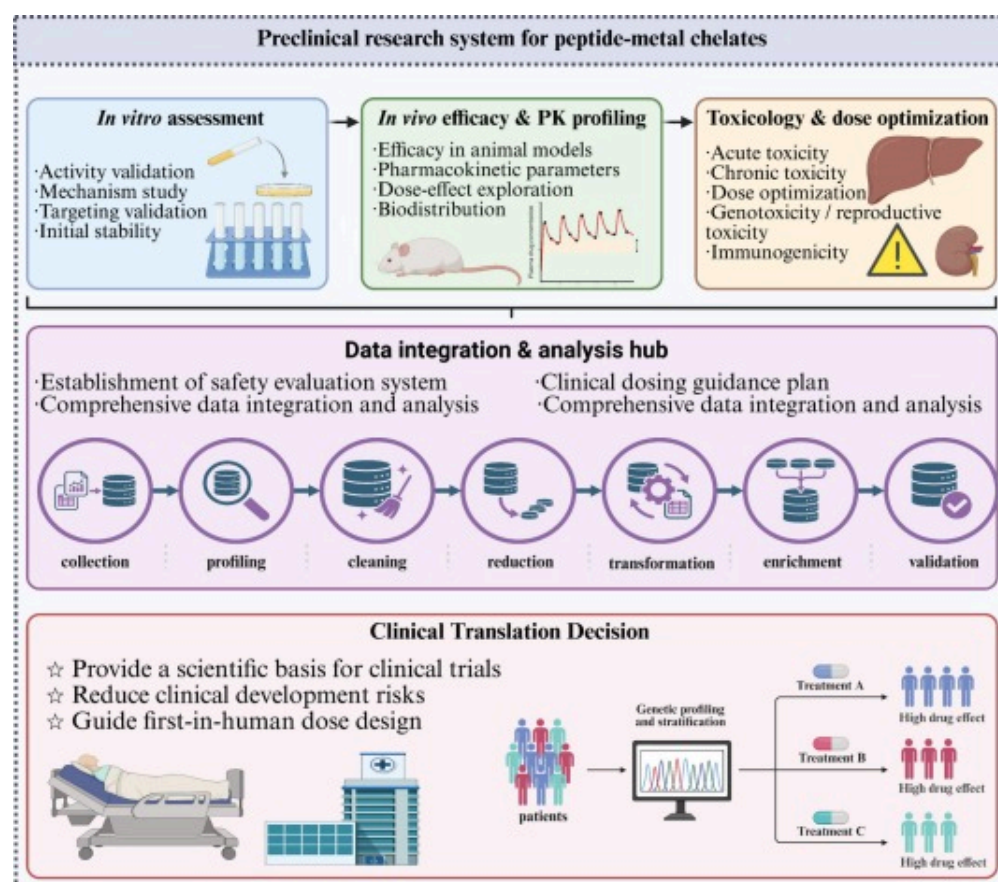
However, the field must also acknowledge the inherent limitations of this platform. Their rapid renal clearance, while minimizing toxicity, may limit tumor accumulation and necessitate frequent dosing or advanced formulation strategies to extend half-life. Additionally, the metal-dependent mechanisms of action introduce unique safety considerations related to metal ion release and long-term tissue accumulation that require careful investigation. Future success will depend on leveraging their unique advantages, such as multifunctionality, synthetic accessibility, and barrier penetration, while addressing these limitations through rational design and advanced delivery systems.

7. Clinical translation and safety assessment

7.1. Lessons from the preclinical battlefield

Preclinical research on peptide-metal chelates constitutes a critical bridge between laboratory discovery and clinical application. As outlined in Fig. 9, this phase follows a

systematic, multi-stage pipeline to comprehensively evaluate candidate compounds. The process typically begins with foundational *in vitro* assessments of activity and mechanism, progresses through *in vivo* efficacy and pharmacokinetic profiling in animal models, and culminates in rigorous toxicological evaluations and dose optimization. It is this integrated workflow that provides the essential scientific foundation for de-risking and informing subsequent clinical trials. However, the standard pipeline often fails to capture the nuanced, real-world complexities that determine translational success or failure. Examining representative cases in which candidates succeeded or failed reveals critical bottlenecks and the strategies that enabled their resolution.



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Fig. 9. Integrated preclinical research system for peptide-metal chelates.

The first critical bottleneck arises from *in vivo* instability, a challenge that has historically limited the use of linear bombesin (BBN) analogs for GRPR-targeted imaging. Early ^{99m}Tc -labeled BBN conjugates suffered rapid enzymatic degradation [241], resulting in poor tumor accumulation. To overcome this, Okarvi and colleagues designed a BBN analog conjugated to methotrexate (MTX-BN) and labeled it with ^{99m}Tc via stannous-tartrate exchange [242]. This conjugate demonstrated good stability against transchelation and high metabolic stability in human plasma. *In vitro* studies confirmed high receptor affinity with significant internalization (19–35%). *In vivo* biodistribution showed tumor uptake of $2.70 \pm 0.44\%$ ID/g at 1 h post-injection, with favorable tumor-to-background ratios, though high kidney uptake (up to 11.05% ID/g) remained a concern for therapeutic applications [242]. More recent advances have incorporated unnatural amino acids at metabolically vulnerable sites; for example, substitution with α -methyl-L-tryptophan in the GRPR antagonist RM2 improved *in vivo* stability, with >30% intact peptide at 30 min versus <20% for unmodified analogs [243]. These findings underscore that enzymatic stability and chelation inertness are the true gatekeepers of *in vivo* success, achievable only through deliberate molecular engineering.

The second bottleneck is enzymatic instability, which limits the therapeutic efficacy of many bioactive peptides [244]. The native tripeptide GHK and its copper complex (GHK-Cu) promote wound healing but are rapidly hydrolyzed by proteases, reducing efficacy. To overcome this, researchers designed Supra GHK-Cu, a supramolecular metallopeptide hydrogel formed by conjugating GHK to a Nap-FF motif that self-assembles upon the addition of Cu^{2+} . Molecular dynamics simulations confirmed that Cu^{2+} bridges peptide molecules, stabilizing the structure. The resulting supramolecular architecture conferred excellent resistance to protease degradation, addressing the key limitation of free GHK-Cu. *In vivo* studies showed that Supra GHK-Cu accelerated wound healing more significantly than GHK-Cu by promoting collagen deposition, angiogenesis, and cell proliferation [245]. This case demonstrates that supramolecular self-assembly can enhance biostability, offering a strategy to overcome enzymatic degradation.

The third bottleneck is the PK/PD disconnect, in which plasma pharmacokinetics fail to predict therapeutic outcomes. This challenge has been masterfully addressed by ^{177}Lu -DOTATATE (Lutathera®), a somatostatin analog conjugate used for peptide receptor radionuclide therapy in neuroendocrine tumors [246]. Early trials revealed that although blood clearance was rapid, tumor retention varied dramatically across patients, and nephrotoxicity emerged as the dose-limiting toxicity with delayed onset poorly correlated with acute pharmacokinetic parameters [247]. The breakthrough came through dosimetry-guided individualized dosing: tailoring administered activity to deliver personalized absorbed doses based on pre-therapy ^{68}Ga -DOTATATE PET/CT reduced renal toxicity while maintaining efficacy. Concurrent infusion of cationic amino acids competitively inhibited renal tubular reabsorption, reducing renal absorbed dose [247]. The cardinal lesson is that for targeted radionuclide therapies, tumor and organ dosimetry is the true determinant of therapeutic index, rather than plasma pharmacokinetics.

These three cases illustrate a common theme: successful preclinical progression is closely associated with early identification and mitigation of potential failure modes. Strategies such as enhancing enzymatic stability, implementing supramolecular stabilization, and addressing PK/PD unpredictability proved most effective when incorporated at the design stage rather than after initial setbacks. The insights gained from these examples inform the discussion of clinical translation challenges that follows.

7.2. Current status of clinical trials

Although clinical trials of peptide-metal chelates are still at a relatively early stage, these compounds have demonstrated significant therapeutic potential in multiple disease areas, particularly in neurodegenerative disorders and osteoporosis, by exhibiting high targeting specificity, favorable bioavailability, and promising safety profiles. This provides robust experimental evidence and strong support for the subsequent clinical translation and application of this innovative therapeutic strategy [248].

Peptide-metal chelates demonstrate significant potential in clinical trials and applied research across the biomedical field, including tumor therapy and neurodegenerative disease intervention, by leveraging the synergistic advantages of peptide-targeted recognition and the biological activity of metal ions. In precision oncology, the peptide-lutetium chelator ^{177}Lu -PSMA-617, which targets prostate-specific membrane antigen, demonstrated in its Phase III VISION study that combination therapy reduces the risk of disease progression or death by 60% and extends overall survival to 15.3 months in patients with PSMA-positive metastatic castration-resistant prostate cancer [249]. In the field of neurodegenerative diseases, the gadolinium chelating peptide Gd-DO3A-VA targets activated microglia in Alzheimer's disease mouse models. It inhibits the NF- κ B inflammatory pathway and A β deposition, significantly improving memory function in mice. This approach offers novel insights into integrated diagnosis and treatment [250]. In the treatment of iron metabolism disorders, porcine skin peptide-iron chelator (PSP-Fe) effectively restores hemoglobin and serum ferritin levels in iron-deficient anemia rat models. It demonstrates superior regulation of the gut microbiota compared to traditional iron supplements and offers significant safety advantages [251]. However, challenges remain in developing such compounds, underscoring the need for dose optimization and control of off-target toxicity for clinical translation.

Ongoing clinical trials of peptide-metal chelates have expanded their indications and the types of drugs they target. Leveraging the synergistic properties of peptide-targeted recognition and metal-ion bioactivity, these compounds demonstrate clear translational potential for the precision treatment of solid tumors, tumor imaging diagnostics, and the intervention for neurodegenerative diseases. The peptide-lutetium-177 chelator ^{177}Lu -FAP-2286, targeting fibroblast activation protein (FAP), is currently in Phase I/II clinical trials. Studies in advanced lung cancer patients demonstrated an objective response rate of 77.78%, with a median progression-free survival of 6 months and a median overall survival of 10 months. No Grade III/IV toxicities were observed, and patients experienced significant improvement in dyspnea and cancer pain symptoms. This offers a novel targeted therapeutic pathway for refractory solid tumors [252]. Gd-DOTA-G3CNGRC, a

gadolinium-chelated peptide targeting CD13, demonstrated high specificity and low toxicity as an MR probe for evaluating anti-angiogenic therapeutic efficacy in Phase I preclinical translational studies with an IC_{50} value of $7.03\mu M$. It enables early assessment of treatment efficacy by monitoring changes in tumor signal intensity, offering a novel direction for the application of peptide-metal complexes in diagnostic and therapeutic support [253].

Peptide-metal chelates demonstrate outstanding clinical efficacy and safety through the synergistic advantages of peptide targeting and the biological activity of metal ions. Related fields are advancing rapidly from proof of concept to practical application. Several peptide-based therapeutics are currently advancing through Phase III clinical trials globally, with indications spanning diverse therapeutic areas, including cancer, diabetes, and COVID-19. Their market sales are growing rapidly, highlighting significant clinical demand and potential. However, the clinical translation of peptide-metal chelates still faces urgent challenges, including drug stability, delivery regimen optimization, and long-term safety.

7.3. Challenges and countermeasures in translational medicine

The transition of peptide-metal chelates from laboratory research to clinical application faces multiple core challenges, including large-scale production, quality control, stability management, optimization of administration regimens, and regulatory approval. These challenges require breakthroughs through coordinated, multidimensional, and systematic strategies.

Scalable production and quality control present the primary bottlenecks. Traditional chemical synthesis processes involve cumbersome steps and are limited to production capacities of 2000–3000l, making it difficult to meet ton-scale market demand. Additionally, treating organic wastewater is costly. Quality control requires the combined use of technologies such as HPLC, CE, and LC-MS to precisely identify missing peptides, broken peptides, and related substances, as well as degradation impurities, ensuring

product purity and safety. Regarding stability, peptides are susceptible to protease hydrolysis and oxidation. Stability can be enhanced through chemical modifications such as replacing L-amino acids with D-amino acids, N-terminal acetylation, or cyclization. Formulation development must avoid extreme pH conditions, high temperatures, and high-humidity environments [254]. Regarding administration routes, traditional injectable formulations exhibit poor patient compliance, prompting the development of sustained-release microspheres, implantable devices, and oral formulations. Sustained-release technologies can effectively reduce enzymatic degradation and extend the therapeutic window [255]. Regulatory approval must account for the structural peculiarities of peptide-metal chelates, referencing relevant technical guidance principles to standardize the entire process from API impurity control and stability assessment to clinical trial design. Industry-academia-research collaboration provides critical support for technological breakthroughs and the translation of research outcomes into practical applications.

Peptide-metal chelates demonstrate promising applications in fields such as oncology and neurodegenerative diseases due to their synergistic advantages of targeting and biological activity. Although challenges in production and stability remain, advancements in synthetic techniques and formulation processes are expected to overcome these bottlenecks. These innovative drugs have the potential to provide safer, more effective treatment options.

7.4. Translational roadmap from Lutathera®

The preceding sections have outlined the multifaceted challenges in translating peptide-metal chelates from bench to bedside. To transform these challenges into actionable guidance, this section distills a translational roadmap using ^{177}Lu -DOTATATE (Lutathera®) as a paradigm. As the first FDA-approved peptide receptor radionuclide therapy (PRRT) agent, its journey from discovery to approval in 2018 offers invaluable insights into the critical stages, requisite data packages, and regulatory hurdles that define successful clinical translation [246].

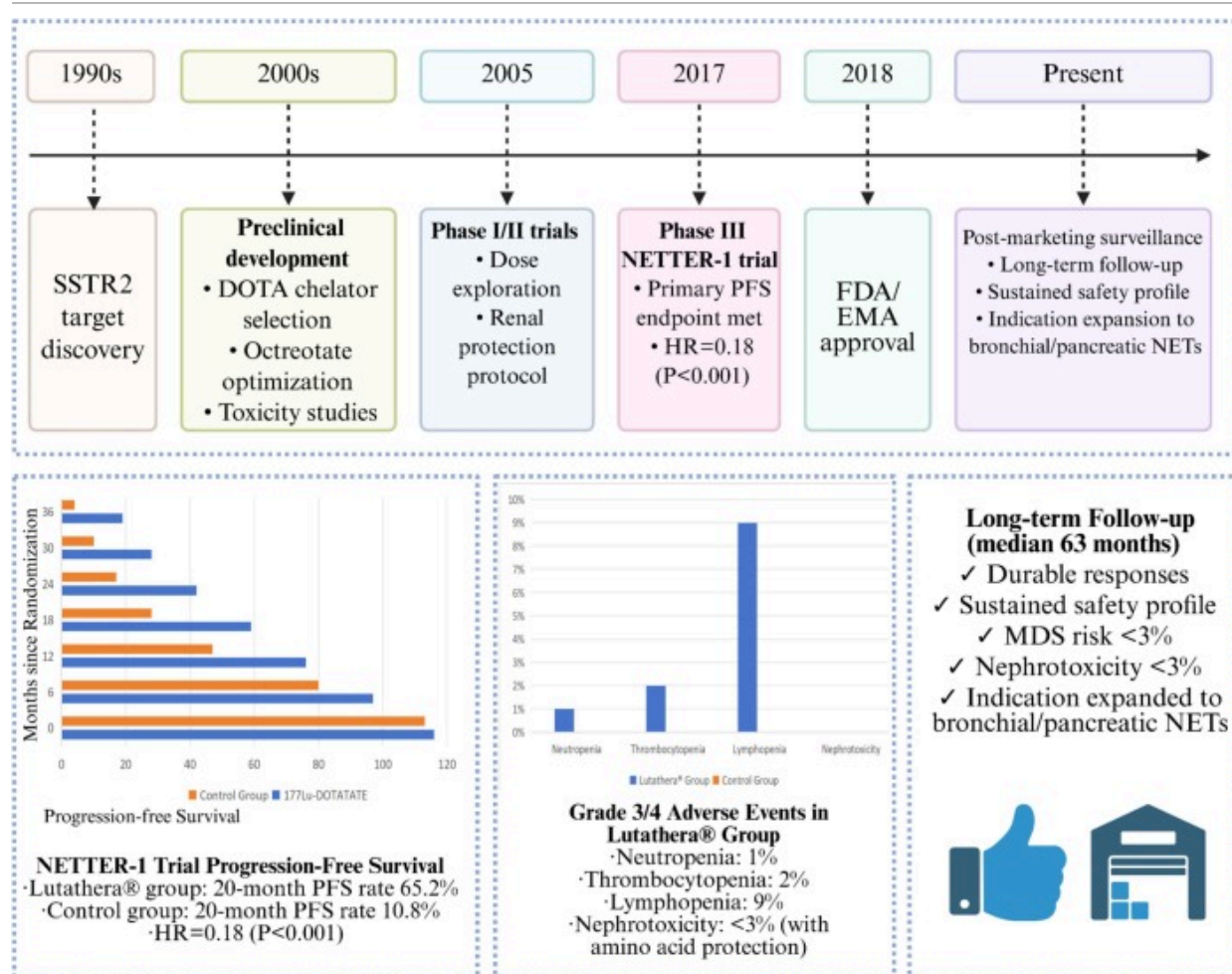
The journey began with early discovery and preclinical development, during which the foundational observation that somatostatin receptors are overexpressed on neuroendocrine tumor cells provided the biological rationale for peptide-based targeting [256]. Early preclinical work focused on optimizing the peptide analog for enhanced receptor affinity and metabolic stability, while selecting the macrocyclic chelator DOTA for its kinetic inertness to prevent premature release of the cytotoxic ^{177}Lu radionuclide [247]. The preclinical data package required at this stage included *in vitro* receptor binding affinity and specificity assays, *in vivo* biodistribution and pharmacokinetic studies in tumor-bearing mouse models to establish tumor uptake and retention kinetics, dosimetry calculations to estimate absorbed radiation doses to tumors and critical organs such as kidneys and bone marrow, and acute and chronic toxicology studies in rodent and non-rodent species to establish the no-observed-adverse-effect level (NOAEL) and identify potential target organs of toxicity [257]. A critical lesson from this stage is that chelation inertness and metabolic stability must be engineered from the outset, not optimized retrospectively.

The subsequent clinical development of Lutathera® followed a phased approach, each generating specific data required for regulatory submission. Phase I trials established the maximum tolerated dose, dose-limiting toxicities, and preliminary dosimetry in patients [258]. A key innovation at this stage was the integration of companion diagnostic imaging using ^{68}Ga -DOTATATE PET/CT to prospectively identify patients with sufficient SSTR2 expression, a paradigm of personalized medicine that significantly enriched the subsequent efficacy trials [259]. The pivotal Phase III NETTER-1 trial randomized patients with progressive midgut neuroendocrine tumors to receive either Lutathera® plus high-dose octreotide or high-dose octreotide alone. The trial met its primary endpoint, demonstrating a hazard ratio for progression-free survival of 0.18 [260]. The safety database, comprising over 300 patients, showed expected and manageable adverse events, primarily hematologic and renal, with no unexpected safety signals. This data package, consisting of a positive pivotal trial, supportive safety data, and an integrated companion diagnostic, served as the basis for FDA and EMA approvals in 2018 [246].

Parallel to clinical development, manufacturing and quality control considerations posed unique challenges for this radiopharmaceutical. The manufacturing process must ensure GMP-compliant synthesis of the peptide-chelator conjugate (DOTATATE), reproducible high-yield radiolabeling with ^{177}Lu at the clinical site or central pharmacy, rigorous quality control testing for radiochemical purity, sterility, and endotoxins within the short half-life of the radionuclide, and stability of the final formulated product during transport and storage. Regulatory expectations for radiopharmaceuticals require validation of the entire manufacturing process, including the radiolabeling procedure performed at end-user sites.

Following approval, post-marketing surveillance and long-term follow-up have continued to shape the clinical utility of Lutathera®. Long-term follow-up from the NETTER-1 trial with a median of 63 months confirmed durable responses and maintained the safety profile, with myelodysplastic syndrome and nephrotoxicity remaining low at less than 3% when renal protective agents such as amino acid infusions are used as established in the clinical protocol [261]. Post-marketing studies have expanded the evidence base to include patients with bronchial and pancreatic neuroendocrine tumors, and combination trials with other agents are ongoing [262]. This stage underscores that regulatory approval is not the endpoint, as continuous safety monitoring and evidence generation are required to maintain approval and expand indications.

As shown in the Fig. 10, this roadmap, distilled from the Lutathera® journey, provides a template for the clinical translation of future peptide-metal chelates. It emphasizes that success requires not only scientific innovation but also strategic planning across preclinical data generation, clinical trial design, manufacturing development, and regulatory engagement. Each stage demands specific expertise and data packages, and integrating these components from the earliest stages of discovery is essential for navigating the complex path from bench to bedside.



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Fig. 10. Lutathera® clinical development and key outcomes.

8. Challenges and future outlook

8.1. Current challenges and bottlenecks

The journey of peptide-metal chelators from concept to clinical application is far from smooth. A series of profound and interrelated scientific and engineering challenges constitute the primary barriers to their translation.

The potential threat of resistance evolution cannot be overlooked. Current mechanisms based on metal uptake or metal-mediated cytotoxicity inherently interfere with physiological processes in pathogens or tumor cells. However, the formidable adaptive capacity of living organisms may foster diverse evasion strategies. For instance, bacteria could “reverse-hijack” metals by downregulating target receptor expression, activating alternative metabolic pathways, or, more intricately, synthesizing endogenous high-affinity chelators, thereby diminishing drug efficacy [263]. This potential adaptive resistance necessitates research that transcends the efficiency of *in vitro* models. Prospective, systematic, and interdisciplinary efforts are required to elucidate its molecular evolutionary pathways, providing enduring and reliable foundations for drug design.

Inherent pharmacokinetic limitations and delivery barriers constitute another major bottleneck. Most peptide-metal chelates exhibit suboptimal *in vivo* behavior due to constraints imposed by their molecular structure and physicochemical properties. Oral administration faces multiple barriers, including gastric acid degradation, protease hydrolysis, and competition with endogenous intestinal metal ions, resulting in extremely low bioavailability. Even with injectable routes, factors such as molecular weight and hydrophilicity often lead to short plasma half-lives and rapid renal clearance. These compounds struggle to penetrate physiological barriers, such as the blood-brain barrier or the dense stroma of solid tumors, making it difficult to achieve and maintain therapeutic concentrations in target tissues [264], [265]. Overcoming these obstacles is crucial for realizing their *in vivo* therapeutic potential [266].

Long-term safety concerns and the “black box” of metal metabolism warrant serious attention. The introduction of exogenous metal ions or radionuclides introduces unique toxicity risks. Metal ions may accumulate ectopically in tissues such as the liver, spleen, or

bones, triggering organ-specific toxicity or long-term functional impairment [267]. More critically, a systematic understanding of the precise metabolic fate, distribution patterns, and ultimate clearance pathways of these therapeutic metal chelates within the complex human environment remains lacking. This metabolic “black box” state makes a comprehensive assessment of their long-term biosafety exceptionally challenging.

The feasibility and stability control of large-scale production represent practical barriers to industrialization. Scaling up from milligram-level synthesis in the laboratory to mass production meeting clinical demand poses significant challenges in process amplification. The chemical stability of peptide-metal complexes is highly dependent on environmental conditions, including pH, ionic strength, and redox state. At the same time, the peptide chains themselves are prone to hydrolysis, oxidation, or conformational change [268]. Establishing a stable, controllable, cost-effective, and GMP-compliant large-scale production process while ensuring end-product stability during storage and transportation represents an engineering challenge that must be overcome for commercialization.

The translational promise of peptide-metal chelates cannot be realized without addressing the practical complexities of manufacturing. While previous sections have focused on biological performance, the transition from laboratory-scale synthesis to commercial production presents multifaceted challenges that must be addressed early in development, including scalable synthetic routes, formulation stability under storage conditions, and compliance with Good Manufacturing Practice (GMP) requirements.

Traditional solid-phase peptide synthesis (SPPS), while efficient for research-scale production, faces limitations in cost-effectiveness and environmental sustainability when scaled to kilogram quantities. The substantial solvent waste, the expense of protected amino acids, and the batch-to-batch variability inherent to SPPS necessitate alternative approaches for commercial manufacturing. Emerging strategies focus on enzymatic and chemoenzymatic synthesis as greener, more scalable alternatives. Recent advances have demonstrated that three-enzyme cascades can efficiently construct metal-chelating

noncanonical amino acids such as bipyridyl-L-alanine (BpyAla) from readily available glycine and aldehyde precursors, with each enzyme produced at high titers exceeding 100mg per liter of culture [269]. This modular chemoenzymatic approach operates with minimal workup and enables access to a range of metal-chelating amino acids that were previously prohibitively expensive. Looking forward, synthetic biology platforms offer transformative potential. Engineering microbial cell factories to produce target peptides from inexpensive carbon sources, followed by controlled metal addition, enables one-pot assembly of final chelate complexes. This approach has demonstrated feasibility for several clinically relevant peptides, with production yields reaching gram-per-liter scales in optimized fermentation processes [268]. Beyond cost reduction, biomanufacturing enables access to structurally complex peptides with stereochemical configurations and post-translational modifications unattainable by chemical synthesis alone.

The stability of peptide-metal chelates during storage and transport represents a critical yet often underappreciated hurdle. These complexes are susceptible to multiple degradation pathways: peptide backbone hydrolysis, oxidation of sensitive residues (methionine, cysteine), metal ion dissociation or transchelation, and conformational changes that alter biological activity. The chelation stability of metal ions directly influences product shelf-life, with macrocyclic chelators (DOTA, NOTA) generally conferring greater kinetic inertness than linear analogs, reducing the risk of metal release during storage. Studies on hemp protein peptide-iron chelates have demonstrated that optimized formulations exhibit high chelation rates across 25–65°C and 4.0–7.0 pH, with stability maintained for 35 days of storage at ambient temperature [270]. Similarly, egg yolk peptide-iron chelates have shown iron release rates maintained at approximately 5% under alkaline conditions, with no significant differences across temperatures from 30°C to 80°C, and substantially lower iron release (18%) during simulated gastrointestinal digestion compared to 97.93% for FeSO₄ [271]. Critical formulation parameters include pH, buffer selection, excipients, and container closure systems that minimize moisture and oxygen exposure. For radiopharmaceutical applications, stability considerations are particularly stringent due to the added complexity of radionuclide decay. The short half-

lives of many diagnostic and therapeutic isotopes require that manufacturing, quality control, and clinical administration occur within tight time windows, necessitating cold-kit formulations that are radiolabeled immediately prior to patient administration.

Transitioning to clinical use mandates adherence to GMP standards, which impose rigorous requirements on every aspect of manufacturing. For peptide-metal chelates, the regulatory framework must accommodate their hybrid nature, which combines elements of synthetic peptides, metal coordination complexes, and in some cases, radiopharmaceuticals. Key GMP considerations include raw material control, process validation, in-process testing, and final product release specifications [192]. Quality control requires a battery of orthogonal analytical methods. High-performance liquid chromatography (HPLC) monitors peptide purity and detects degradation products such as truncated or oxidized sequences. Mass spectrometry confirms molecular identity and coordination stoichiometry. Inductively coupled plasma mass spectrometry (ICP-MS) quantifies metal content and detects potential contamination. For chiral peptides, circular dichroism or chiral HPLC ensures the correct stereochemistry, which directly affects metal-binding affinity and biological activity [192]. Importantly, batch-to-batch consistency, which is critical to scalable commercialization, demands tight control over peptide profiles and coordination chemistry. Platform-led development and standardized peptide generation enable reproducible manufacturing. For radiopharmaceuticals, GMP requirements extend to the radiolabeling process, which is often performed at clinical sites or centralized pharmacies rather than at the primary manufacturing facility, necessitating validated procedures, clear acceptance criteria, and robust training programs to ensure consistent product quality across multiple end-user sites.

The manufacturing realities outlined above carry important implications for translational strategy. First, scalability must be considered early because processes designed for laboratory-scale synthesis may not translate directly to commercial production, and late-stage process changes can trigger additional regulatory review. Second, stability should be engineered into the molecular design, with macrocyclic chelators and stabilized peptide backbones (D-amino acids, cyclization) offering advantages from both biological and

manufacturing perspectives. Third, analytical method development should parallel product development, ensuring that appropriate quality attributes can be measured and controlled before pivotal studies begin. Collaboration between peptide suppliers, formulation scientists, and regulatory experts is essential to navigate these complexities. The field is moving toward integrated development models where manufacturing considerations inform molecular design from the outset, rather than being addressed retrospectively after clinical proof-of-concept is established.

8.2. Key directions for technological innovation

The translation of peptide-metal chelates from laboratory concepts to clinical reality hinges on overcoming key technological barriers, with feasibility and stability in large-scale production posing primary industrialization hurdles. Scaling up from lab synthesis to mass production remains challenging, as peptide-metal complex stability depends critically on pH, ionic strength, and redox state. At the same time, peptides themselves are prone to degradation. Establishing a stable, cost-effective, GMP-compliant process and ensuring product stability during long-term storage and transport remain critical engineering obstacles to commercialization. Addressing these challenges requires innovation across multiple fronts, from molecular design to delivery systems and manufacturing paradigms.

At the molecular design level, rational design and artificial intelligence tools are emerging as core drivers of accelerated discovery. Rather than relying on traditional trial-and-error screening, researchers now leverage computer-aided design and AI-powered prediction platforms to perform virtual screening and peptide sequence optimization at unprecedented speed. Machine learning models trained on existing peptide-metal coordination data can predict binding affinity, selectivity, and coordination kinetics for novel sequences, enabling the design of more stable and efficient candidate molecules from the outset. For instance, deep learning algorithms have been successfully applied to predict metal-binding motifs in peptide sequences, achieving over 85% accuracy in identifying copper and zinc coordination sites [272]. More advanced approaches employ

generative adversarial networks (GANs) to propose entirely novel peptide sequences optimized for specific metal ions, with subsequent experimental validation confirming that AI-designed candidates exhibit 2–3-fold higher metal-binding affinity than library-derived sequences. Concurrently, chemical biology tools, such as enzymatic synthesis, especially *via* the macrocyclizing enzyme PCY1, provide efficient and green pathways for constructing structurally complex macrocyclic peptide chelators with precise stereochemical configurations that cannot be achieved by conventional solid-phase synthesis [268], [273]. Furthermore, designing self-assembling peptide sequences driven by metal-ion coordination to form ordered nanostructures not only enables their use as drug carriers but also endows the assemblies themselves with novel biological functions, such as enzyme-mimetic catalytic activity or antimicrobial properties [274].

The development of smart delivery systems aims to overcome the persistent bioavailability bottlenecks that have historically limited the efficacy of peptide therapeutics. Current research focuses on developing nanocarriers that are responsive to specific disease microenvironment characteristics. For example, nanoparticles based on biodegradable metal-organic frameworks (MOFs) can be engineered with acid-labile linkers that trigger cargo release specifically in tumor microenvironments characterized by low pH (pH6.5–6.8, compared with physiological pH7.4). Similarly, polymeric nanoparticles incorporating thioketal linkages respond to elevated reactive oxygen species (ROS) in inflamed or malignant tissues, achieving site-specific payload release with minimal systemic exposure [275]. Beyond simple responsive release, more sophisticated approaches involve developing biomimetic or biohybrid delivery systems. One promising strategy combines functional peptides with cell membrane vesicles derived from red blood cells or mesenchymal stem cells, which confer immune evasion and prolonged circulation half-life [276]. Metal-coordinated dendrimers represent another elegant solution, in which branched polymers coordinate peptide-metal payloads through reversible metal-ligand interactions, enabling efficient intracellular delivery with enhanced transmembrane permeability and cytoplasmic bioavailability [277], [278]. The bispecific antibody pre-targeting strategy represents a particularly transformative

approach for radiotheranostic applications [279]. In this strategy, patients first receive a bispecific antibody that binds both tumor-associated antigens and a radiolabeled peptide. After sufficient time is allowed for tumor accumulation and clearance of unbound antibody from circulation, a small, rapidly diffusing radiolabeled peptide chelate is administered. This peptide selectively binds to the pre-localized antibody, dramatically improving tumor-to-background ratios while minimizing radiation exposure to healthy tissues.

Biosynthetic and green manufacturing technologies are poised to revolutionize production paradigms for peptide-metal chelates. Rather than relying on resource-intensive solid-phase peptide synthesis, which generates significant solvent waste and requires expensive protected amino acids, researchers are engineering microbial cell factories capable of producing target peptides from simple, inexpensive carbon sources [280]. Through synthetic biology, metabolic pathways can be reprogrammed to direct flux toward desired peptide sequences, with subsequent addition of metal ions enabling one-pot assembly of the final chelate complex. This approach has already demonstrated feasibility for several clinically relevant peptides, with production yields reaching gram-per-liter scales in optimized fermentation processes [268]. Beyond cost reduction, biomanufacturing enables access to structurally complex natural product-like peptides with stereochemical configurations and post-translational modifications that cannot be replicated by chemical synthesis, opening new chemical space for the discovery of metal chelators [266]. For instance, ribosomally synthesized and post-translationally modified peptides (RiPPs) from bacterial genomes represent an untapped reservoir of metal-chelating sequences [281], with genome mining revealing hundreds of previously uncharacterized candidates awaiting functional exploration.

Novel formulations and combination therapy strategies directly address the dual imperatives of enhancing efficacy while minimizing toxicity. Chemical modifications such as polyethylene glycolation (PEGylation) remain valuable for improving pharmacokinetic properties, with site-specific PEGylation at non-critical residues extending the circulation half-life from minutes to hours while preserving receptor-binding affinity [275]. However,

the most transformative advances lie in integrating peptide-metal chelates into multi-mechanistic combination therapy frameworks. For oncology applications, co-loading of peptide-metal complexes with chemotherapeutic agents and immunomodulators within a single nanoparticle enables a coordinated attack. In this approach, the metal complex exerts direct cytotoxicity or enables imaging, the chemotherapy agent disrupts tumor proliferation, and the immunomodulator reverses immune suppression, together achieving synergistic effects that overcome resistance mechanisms inherent to single-agent strategies. Specific integration strategies include developing personalized nanodrugs in which targeting peptides direct the delivery of siRNA payloads to silence drug-resistant genes [282], followed by metal-based therapy to eliminate the now-sensitized tumor cells [283], [284]. For infectious disease applications, combining antimicrobial peptides with metal ions in a single chelate exploits dual mechanisms of membrane disruption and metal-induced oxidative stress, making resistance development significantly more challenging for bacteria. The emergence of smart biomaterials has further expanded application scope beyond traditional therapeutics: water vapor-responsive metal-peptide nanofibers that undergo reversible structural changes upon humidity fluctuations have demonstrated utility in wearable sensing platforms for real-time health monitoring. In contrast, metal-peptide hydrogels with shear-thinning properties enable injectable depot formulations for sustained local drug delivery [274].

Collectively, these technological innovations, including AI-guided molecular design, microenvironment-responsive delivery, biosynthetic manufacturing, and rationally designed combination regimens, are converging to address the longstanding barriers that have impeded the translation of peptide-metal chelates. The field is moving from serendipitous discovery toward rational engineering, where candidate molecules are not merely identified but designed from first principles with clinical success as the explicit objective.

8.3. Broad clinical application prospects

Despite formidable challenges, peptide-metal chelates have outlined a clear and compelling application blueprint across multiple cutting-edge fields of precision medicine, leveraging their irreplaceable unique advantages.

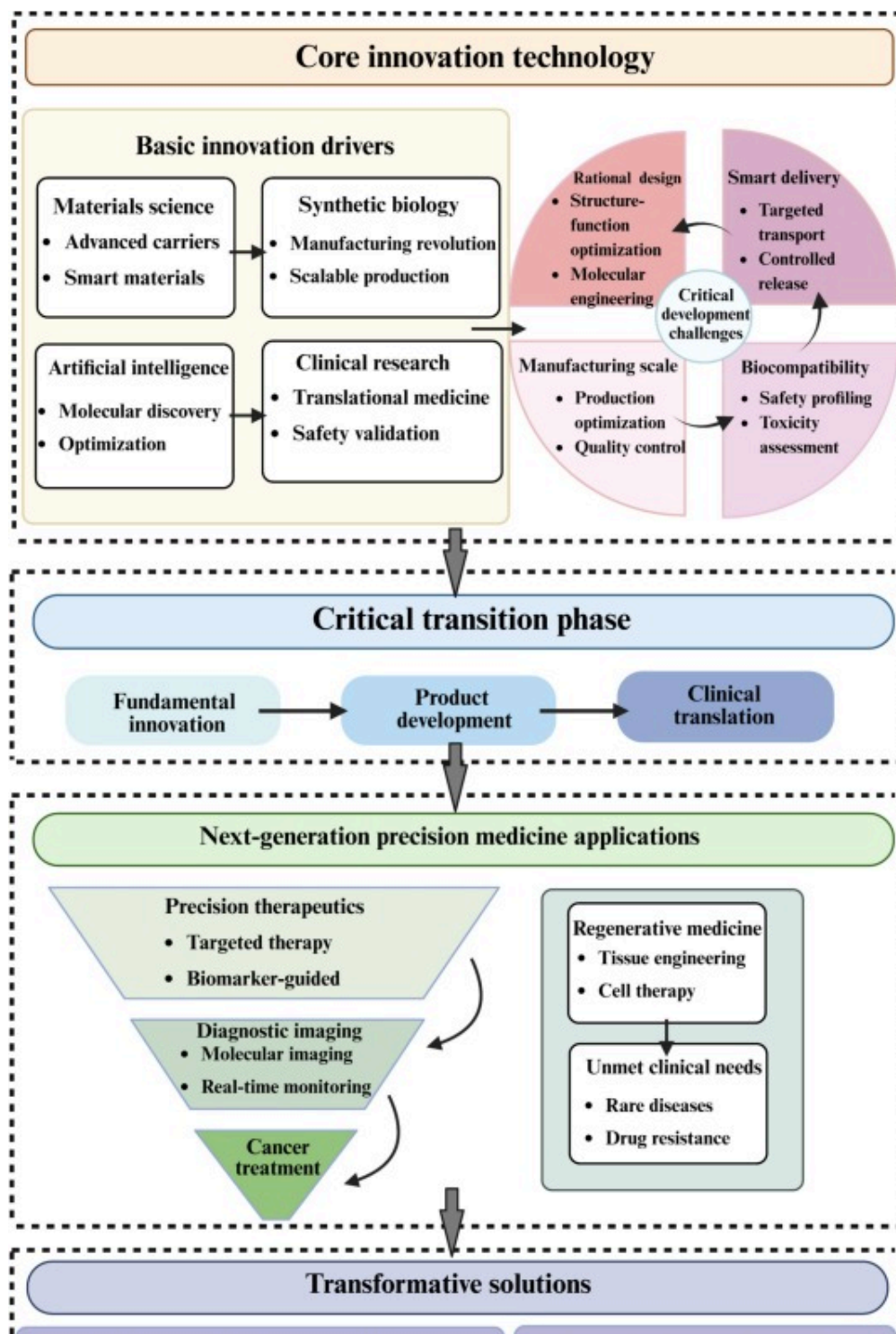
In the field of anti-infective therapy, this strategy has achieved a landmark breakthrough. Iron-carrier-antibiotic conjugates, exemplified by ceftizoxime, successfully “disguise” antibiotics as iron, which is an essential element for bacteria. By exploiting the bacteria's own iron uptake system, these conjugates are actively “ingested” into cells, thereby ingeniously bypassing the formidable outer membrane barrier and efflux pump mechanisms of Gram-negative bacteria [285]. The successful commercialization and application of these drugs not only provide a powerful tool for treating complex infections caused by carbapenem-resistant bacteria but also clinically validate the feasibility of the core concept of “metal-mediated targeting.”

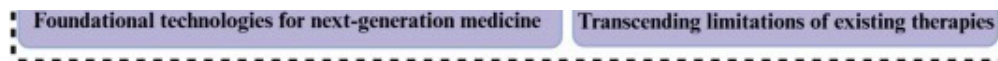
The prospects for integrated tumor diagnosis and therapy are particularly promising. On the one hand, the precise labeling of targeting peptides with radioactive metal nuclides has led to the development of revolutionary dual-purpose diagnostic-therapeutic drugs. For instance, the lutetium-177-labeled PSMA-617 drug enables simultaneous precise imaging of prostate cancer lesions and targeted internal radiation therapy, marking the dawn of a new era in cancer treatment where “seeing is treating” [286]. On the other hand, combining cytokine mimetic or checkpoint inhibitor peptides with immune-adjuvant metal ions such as manganese and copper is being explored to develop next-generation tumor immunotherapies that activate anti-tumor immune responses *in situ* [283]. Peptide-metal self-assembled nanostructures also demonstrate unique value in enhancing drug endosomal escape and in achieving integrated diagnosis and therapy [172].

This platform also exhibits unique potential in emerging fields such as neurodegenerative diseases. Designing peptides capable of crossing the blood-brain barrier and selectively chelating excess free copper and zinc ions in pathological states is considered an effective strategy to intervene in metal homeostasis imbalances and associated protein misfolding

in diseases like Alzheimer's disease, offering novel approaches for such intractable conditions [287]. This highlights the platform's potential to address delivery challenges for central nervous system disorders.

The field of peptide-metal chelates has entered a critical phase, transitioning from fundamental innovation to product development. Future success will heavily depend on deep interdisciplinary integration, such as leveraging materials science for advanced carriers and smart materials [274]. Synthetic biology revolutionizes manufacturing paradigms [268], artificial intelligence accelerates molecular discovery and optimization, while rigorous translational medicine and clinical research remain integral throughout to validate clinical value and safety. As illustrated in Fig. 11, through collaborative innovation, systematically addressing bottlenecks spanning rational design [273], smart delivery [275], [278], biocompatibility [284], and scalable production, peptide-metal chelates hold promise to truly transcend the limitations of existing therapies. They are poised to become foundational technologies for next-generation precision medicine, delivering transformative solutions to numerous unmet clinical needs.





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Fig. 11. Multidisciplinary Integration for Clinical Translation.

To accelerate clinical translation, five priority research directions emerge from this review. First, the development of AI-driven design platforms for predicting metal-binding motifs and optimizing peptide sequences, reducing reliance on trial-and-error screening and enabling rational design of stable, high-affinity chelates from the outset. Second, the engineering of smart, multi-responsive delivery systems that integrate sensing of multiple microenvironmental cues (pH, enzymes, redox) to achieve precise spatiotemporal control of drug release, addressing the persistent bioavailability bottleneck. Third, the advancement of biosynthetic manufacturing technologies, particularly microbial cell factories and enzymatic synthesis platforms, to enable cost-effective, scalable, and environmentally sustainable production of complex peptide-metal chelates. Fourth, the systematic investigation of metal metabolism and long-term safety through integrated pharmacokinetic-toxicodynamic studies, establishing predictive biomarkers for early detection of organ-specific toxicity and enabling evidence-based dose optimization. Fifth, the design of combination therapy regimens that leverage the unique mechanisms of peptide-metal chelates to synergize with immunomodulators, chemotherapeutics, or physical modalities (photothermal, sonodynamic), thereby overcoming resistance and enhancing therapeutic efficacy.

Crucially, none of these directions can be pursued in isolation. Realizing the full potential of peptide-metal chelates demands deep, systematic cross-disciplinary collaboration integrating synthetic chemistry, structural biology, computational modeling, materials science, pharmaceutical sciences, and clinical medicine. Only through such integrated efforts can the field transition from serendipitous discovery to rational engineering, delivering transformative solutions to unmet clinical needs.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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